MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, D-TYPE, DIGITAL, KEY CODE BMD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This document establishes the requirements and test procedures for the acquisition of an electronic D-type flip-flop module.

1.2 Classification. This specification covers two classes of modules

Class I and class II as specified in MIL-STD-1389

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.205 failures per million operating hours for class I or 2.167 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirement specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BMD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/256-1 for class I and M28787/256-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3.0 to 18.0 V $\pm$5%

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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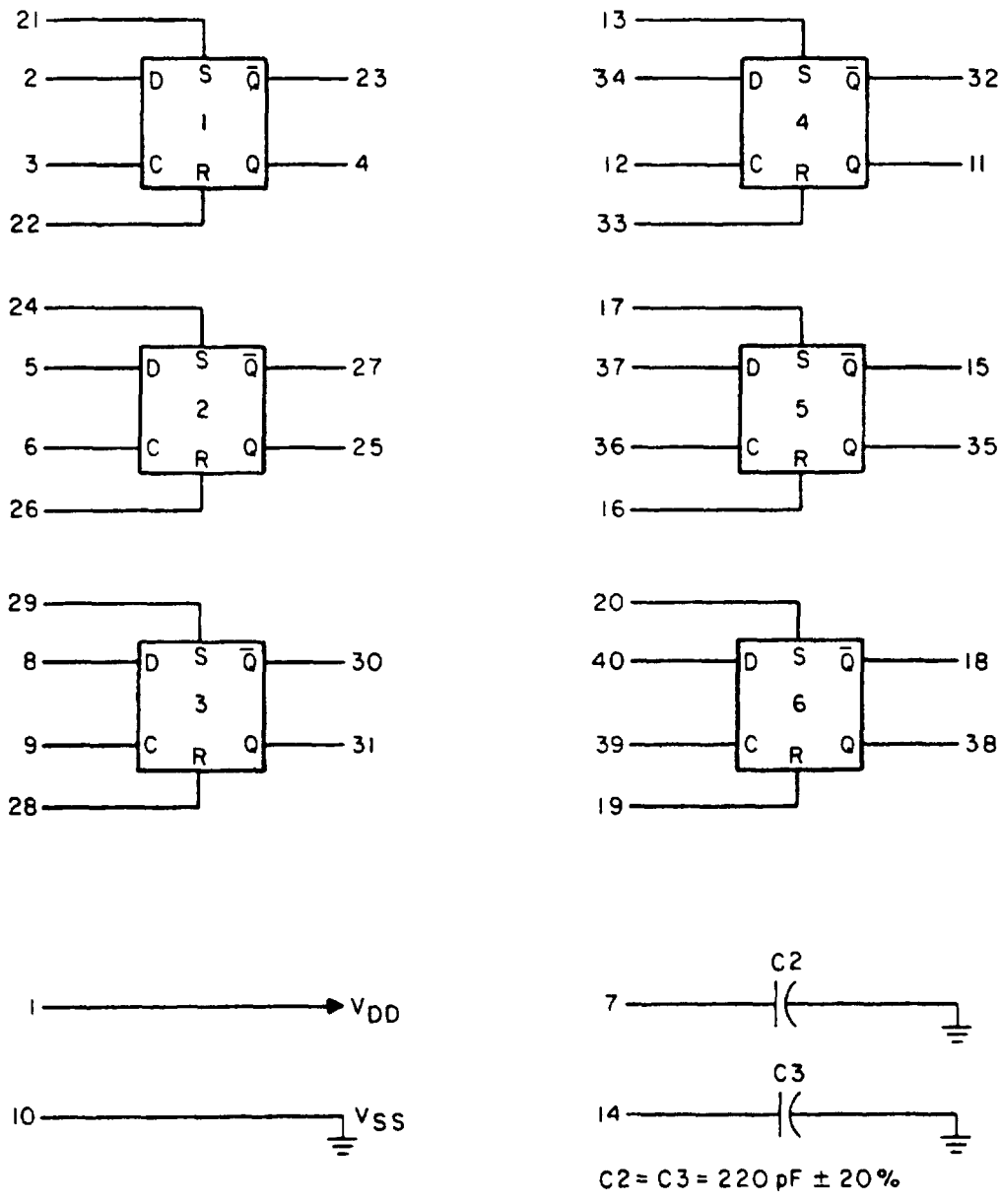


FIGURE 1 BMD module function diagram, D-type flip-flop.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Static output voltage, low level, I_{OL} = 0	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
E1-E12	Static output voltage, high level, I_{OH} = 0	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F12	Static output voltage, low level, V_{DD} = 5 V	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0 0	0.4	V
G1-G12	Static output voltage, high level, V_{DD} = 5 V	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H12	Static output voltage, low level, V_{DD} = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J12	Static output voltage, high level, V_{DD} = 15 V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
K1-K24	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
L1-L24	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1	Static power supply current, low level	XIV	X	0	60	0	60	0	590	0	60	0	1200	µA
M2	Static power supply current, high level	XIV	X	0	60	0	60	0	590	0	60	0	1200	µA

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XV	S	36	360	36	360	40	400	27	275	44	440	ns
N13-N24	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XV	Q	84	840	84	840	93	930	64	640	103	1030	ns
P1-P12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XVI	S	20	200	20	200	22	220	15	150	24	245	ns
P13-P24	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	68	680	68	680	75	750	51	515	83	835	ns
Q1-Q12	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	35	350	35	350	38	385	26	265	43	430	ns
Q13-Q24	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	55	550	55	550	61	610	42	420	67	675	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XVIII	S	35	350	35	350	38	385	26	265	43	430	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XVIII	Q	55	550	55	550	61	610	42	420	67	675	ns
S1-S6	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XIX	S	35	350	35	350	38	385	26	265	43	430	ns
S7-S12	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XIX	S	45	450	45	450	50	500	34	340	55	550	ns
S13-S18	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XIX	Q	55	550	55	550	61	610	42	420	67	675	ns

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S19-S24	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XIX	Q	65	650	65	650	72	720	49	495	79	795	ns
T1-T6	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XX	S	35	350	35	350	38	385	26	265	43	430	ns
T7-T12	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XX	S	45	450	45	450	50	500	34	340	55	550	ns
T13-T18	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	55	550	55	550	61	610	42	420	67	675	ns
T19-T24	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	65	650	65	650	72	720	49	495	79	795	ns
U1-U12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XXI	Q	18	180	18	180	20	200	13	135	22	220	ns
V1-V12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XXII	Q	10	100	10	100	11	110	7	75	12	125	ns
W1-W12	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	15	150	15	150	16	165	11	115	18	185	ns
X1-X12	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIV	Q	15	150	15	150	16	165	11	115	18	185	ns

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Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y6	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXV	Q	15	150	15	150	16	165	11	115	18	185	ns
Y7-Y12	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXV	Q	20	200	20	200	22	220	15	150	24	245	ns
Z1-Z6	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVI	Q	15	150	15	150	16	165	11	115	18	185	ns
Z7-Z12	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVI	Q	20	200	20	200	22	220	15	150	24	245	ns
AA1-AA12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXVII	Q	13	130	13	130	14	145	10	100	16	160	ns
BB1-BB12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXVIII	Q	8	80	8	80	9	90	6	60	10	100	ns
CC1-CC12	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIX	Q	10	100	10	100	11	110	7	75	12	125	ns
DD1-DD12	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	10	100	10	100	11	110	7	75	12	125	ns
EE1-EE6	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXI	Q	10	100	10	100	11	110	7	75	12	125	ns
EE7-EE12	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXI	Q	15	150	15	150	16	165	11	115	18	185	ns

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Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
FF1-FF6	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXII	Q	10	100	10	100	11	110	7	75	12	125	ns
FF7-FF12	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXII	Q	15	150	15	150	16	165	11	115	18	185	ns
GG1-GG12	Maximum shift frequency check	XXXIII	S	900	1100	900	1100	900	1100	900	1100	900	1100	ns
HH	Power supply filter capacitance	XXXIV	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	uF
JJ1, JJ2	C2 and C3 capacitance	XXXV	X	176	264	176	264	176	264	176	264	176	264	pF
KK1-KK24	Input capacitance	XXXVI	Q	2	15	2	15	2	15	2	15	2	15	pF
LL1-LL3	Functional test	XXXVII	X	← See table XXXVII →										
MM1-MM12	Data pulse set time (t_{SLH}), V_{DD} = 5 V, C_L = 50 pF	XXXVIII	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	us
MM13-MM24	Data pulse set time (t_{SHL}), V_{DD} = 5 V, C_L = 50 pF	XXXVIII	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	us
NN1-NN12	Data pulse hold time (t_{HHL}), V_{DD} = 5 V, C_L = 50 pF	XXXIX	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4 4	us
NN13-NN24	Data pulse hold time (t_{HLH}), V_{DD} = 5 V, C_L = 50 pF	XXXIX	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	us
PP1-PP12	Data pulse set time (t_{SLH}), V_{DD} = 10 V, C_L = 50 pF	XL	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	us
PP13-PP24	Data pulse set time (t_{SHL}), V_{DD} = 10 V, C_L = 50 pF	XL	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4 4	3.6	4.4	us

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
QQ1-QQ12	Data pulse hold time (t_{HHL}), V_{DD} = 10 V, C_L = 50 pF	XLI	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µs
QQ13-QQ24	Data pulse hold time (t_{HLH}), V_{DD} = 10 V, C_L = 50 pF	XLI	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µs
RR1-RR12	Data pulse set time (t_{SLH}), V_{DD} = 15 V, C_L = 50 pF	XLII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µs
RR13-RR24	Data pulse set time (t_{SHL}), V_{DD} = 15 V, C_L = 50 pF	XLII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µs
SS1-SS12	Data pulse hold time (t_{HLH}), V_{DD} = 15 V, C_L = 50 pF	XLIII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µs
SS13-SS24	Data pulse hold time (t_{HHL}), V_{DD} = 15 V, C_L = 50 pF	XLIII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through SS shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through SS in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through J, L through JJ, and LL through SS shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions K and L shall be performed at 25°, 60° and 100°C only using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition HH, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through J, L through GG, JJ and LL through SS shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through SS shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

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	Class I	Class II	Unit
Maximum power dissipation	18 max	18 max	mW
Input voltage			
Low level (V_{DD} = 5 V)	0 min, 1.5 max	0 min 1.5 max	V
Low level (V_{DD} = 15 V)	0 min, 4.0 max	0 min, 4.0 max	V
High level (V_{DD} = 5 V)	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
High level (V_{DD} = 15 V)	11.0 min, V_{DD} max	11.0 min, V_{DD} max	V
Input current			
Low level (V_{IL} = 0 V, V_{DD} = 18 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V, V_{DD} = 18 V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5 V)			
Low level (V_{OL} = 0.4 V)			
I_{OL} at -55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
High level (V_{OH} = 4 6 V)			
I_{OH} at -55°C	---	-0.64 min	mA
0° and 25°C	-0.51 min	-0.51 min	mA
60°C	-0.44 min	-0.44 min	mA
100°C	---	-0.36 min	mA
Clock input, V_{DD} = 5 V			
Frequency	2.0 max	2.0 max	MHz
Pulse width	250 min	250 min	ns
Set, reset pulse width	250 min	250 min	ns

6.5 Module function. The D-type flip-flop module containes six independent D-type flip-flops and two 220 pF capacitors with one side connected to V_{SS} for external use.

6.5.1 Input to output relationship. Each gate shall have the input to output relationship of table II for positive logic.

TABLE II. Input output relationship for each module gate.

Inputs				Outputs	
Clock	D	R	S	Q	$\overline{Q}$
↑	L	L	L	L	H
↑	H	L	L	H	L
↓	X	L	L	Q (no change)	$\overline{Q}$ (no change)
X	X	h	L	L	H
X	X	L	H	H	L
X	X	H	H	H	H

1/ H = High voltage level
2/ L = Low voltage level
3/ X = Don't care



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6.5.2 Paralleling of outputs. Paralleling of outputs to obtain extended drive and sink capability is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are

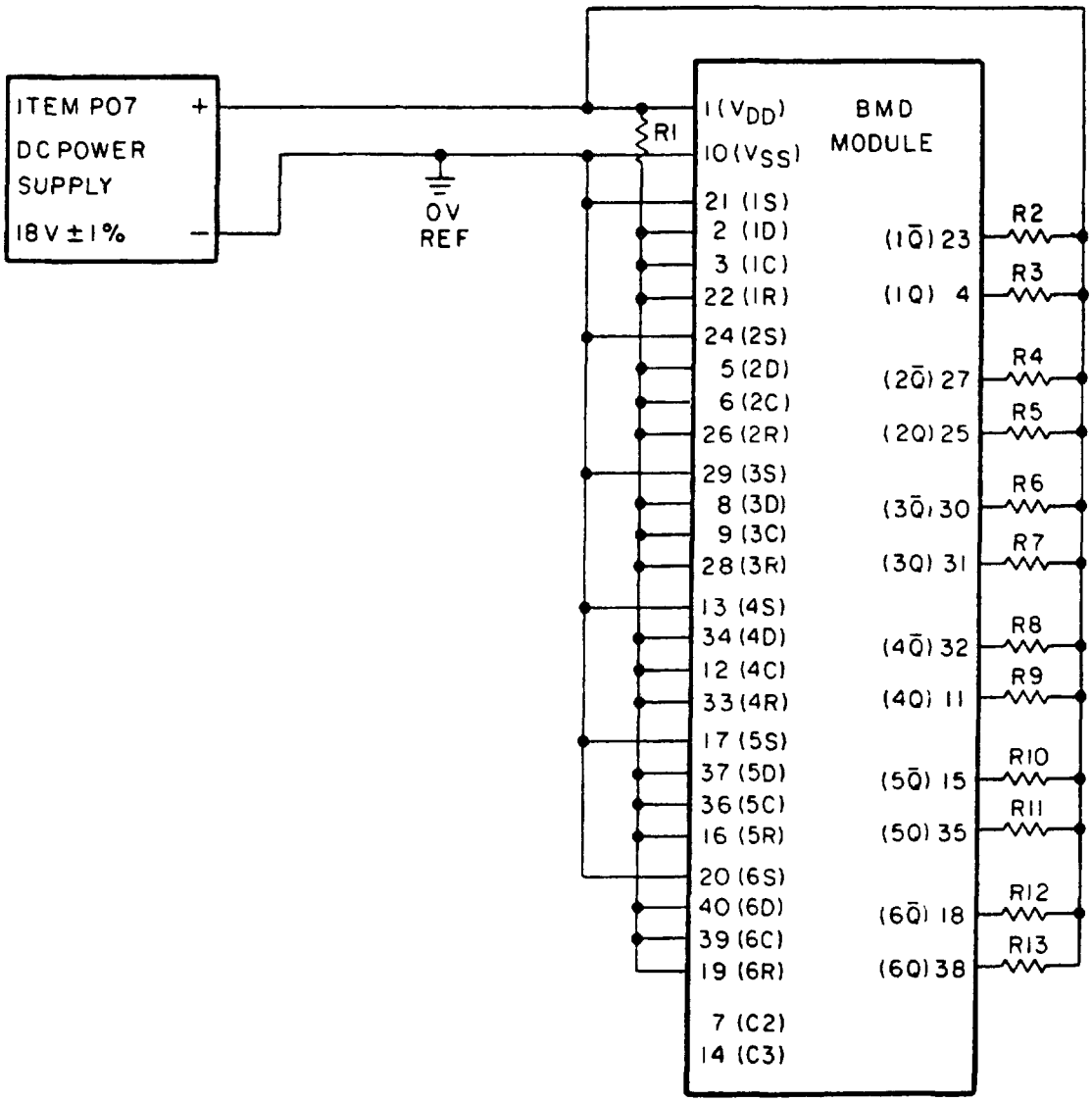
a. 1 and 2.
b. 3 and 4.
c. 5 and 6.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N203)

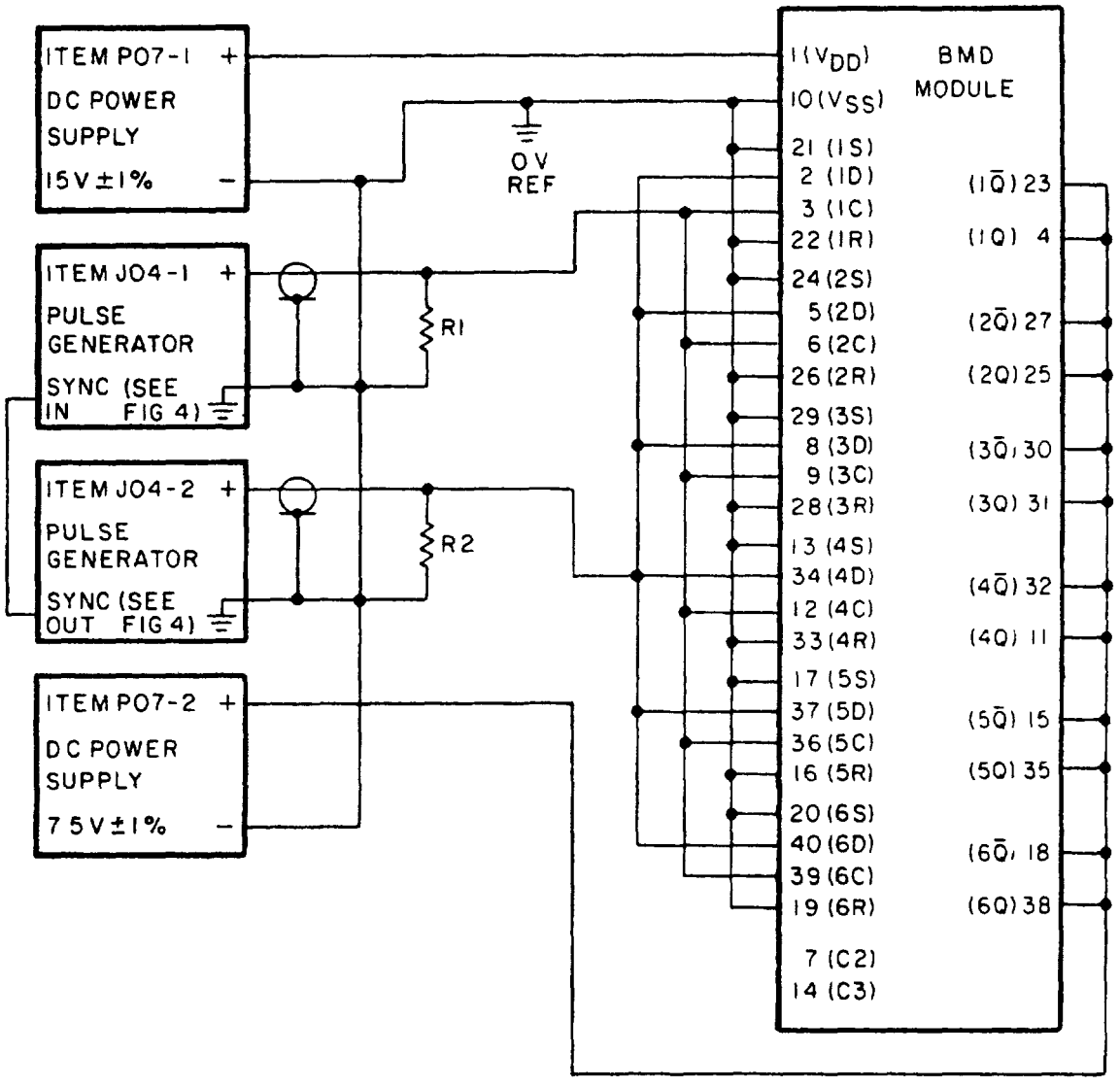
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NOTES
1 R1 = 1 5 kΩ ±5%, 1/4 W
2 R2, R4, R6, R8, R10, R12 = 1 37 kΩ ±1%, 1/8 W.
3 R3, R5, R7, R9, R11, R13 = 6 81 kΩ ±1%, 1/8 W.

FIGURE 2 Operating temperature and life test circuit.

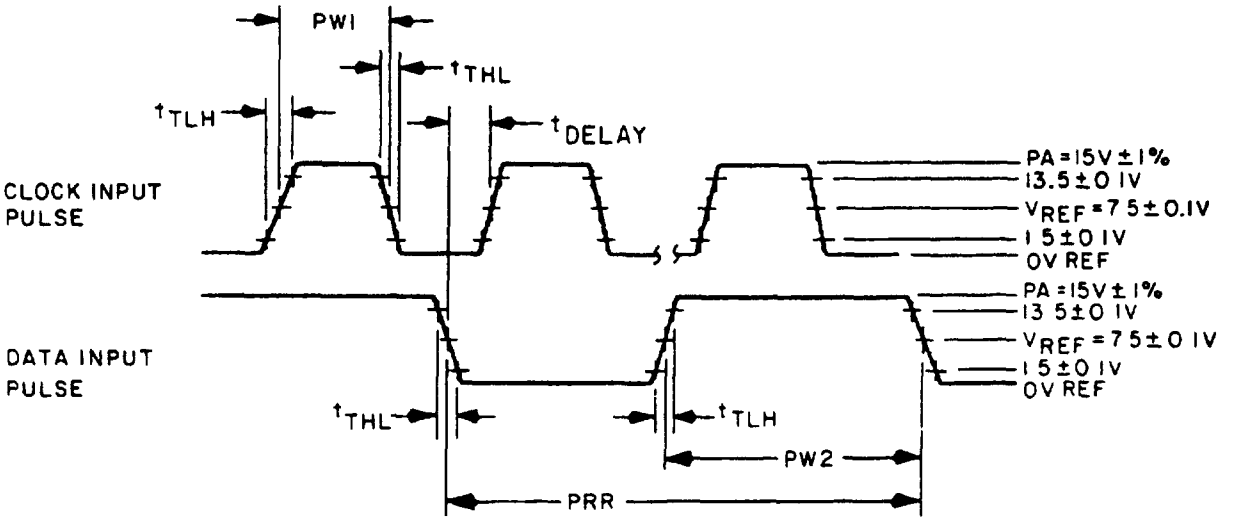
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NOTE: R1 = R2 = 51.1Ω ±5%, 5 W

FIGURE 3 Component temperature test circuit

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NOTES:
1. Item J04-1 parameters (clock input pulse)
   PW-1 = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns
   Double pulse mode
2. Item J04-2 parameters (data input pulse)
   PW2 = 1.0 ±0.2 µs
   $t_{TLH} = t_{THL}$ = 50 ±10 ns
   PRR = 2.0 µs ±5%
3. t_{DELAY} = 250 ±10 ns

FIGURE 4 Waveform setup (component temperature test) for figure 3.

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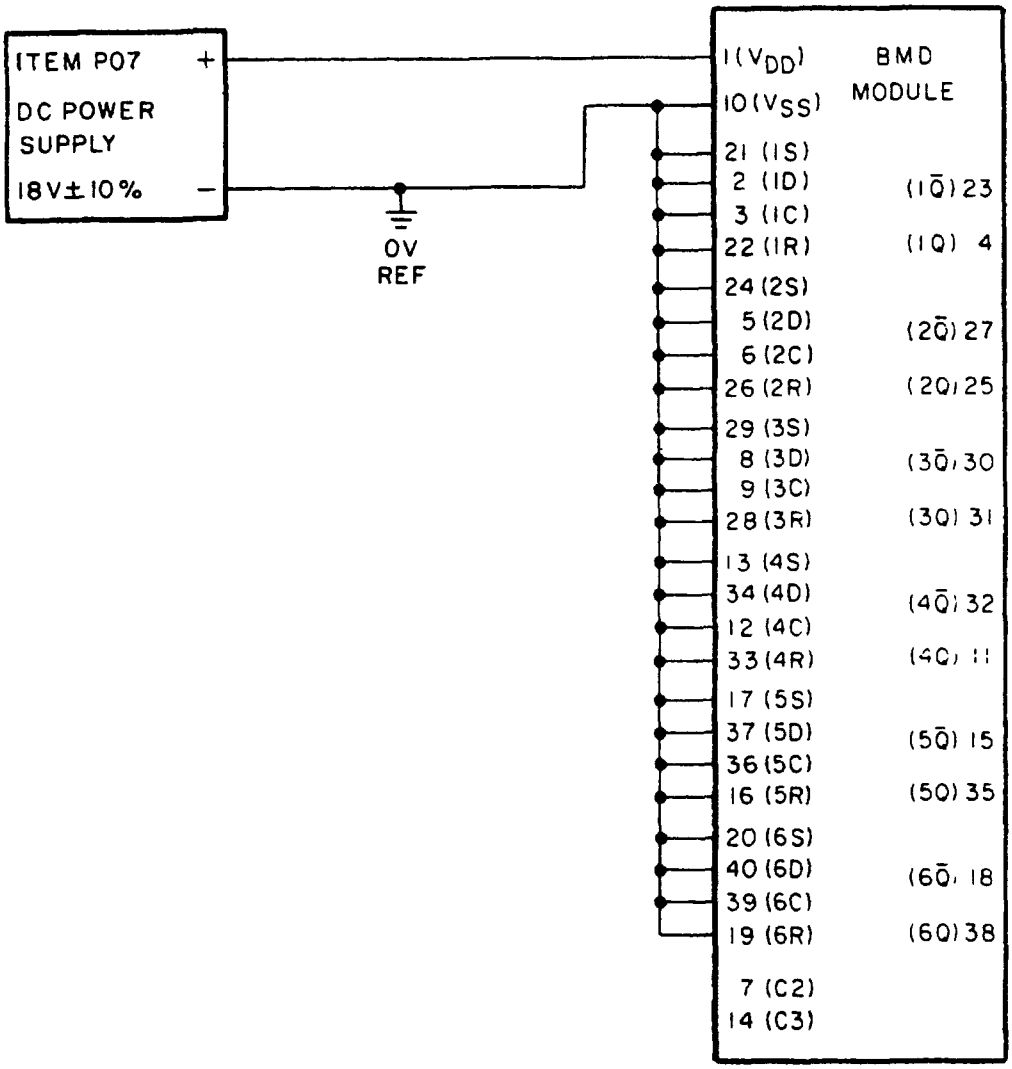


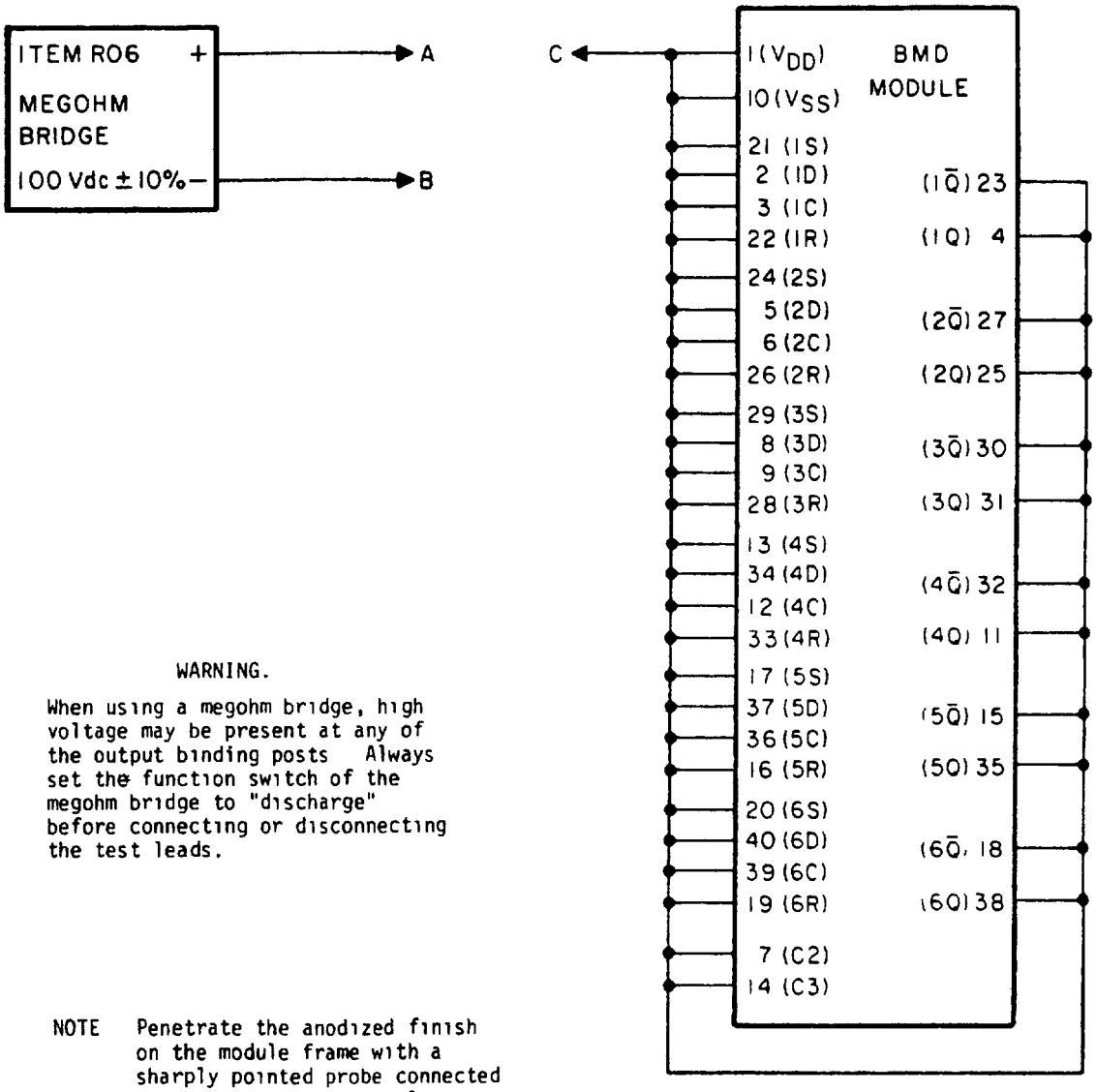
FIGURE 5. Vibration test circuit

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	-	MΩ	6

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WARNING.

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6 Isolation test circuit for table III

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TABLE IV. Input clamp diode voltage (negative) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1S Input clamp diode voltage, negative	21	Open	-0.4	-6.0	V	7
B2	1D	2					
B3	1C	3					
B4	1R	22					
B5	2S	24					
B6	2D	5					
B7	2C	6					
B8	2R	26					
B9	3S	29					
B10	3D	8					
B11	3C	9					
B12	3R	28					
B13	4S	13					
B14	4D	34					
B15	4C	12					
B16	4R	33					
B17	5S	17					
B18	5D	37					
B19	5C	36					
B20	5R	16					

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TABLE IV. Input clamp diode voltage (negative) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B21	6S Input clamp diode voltage, negative	20	Open	-0.4	-6.0	V	7
B22	6D	40					
B23	6C	39					
B24	6R	19					

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TABLE V. Input clamp diode voltage (positive) test.

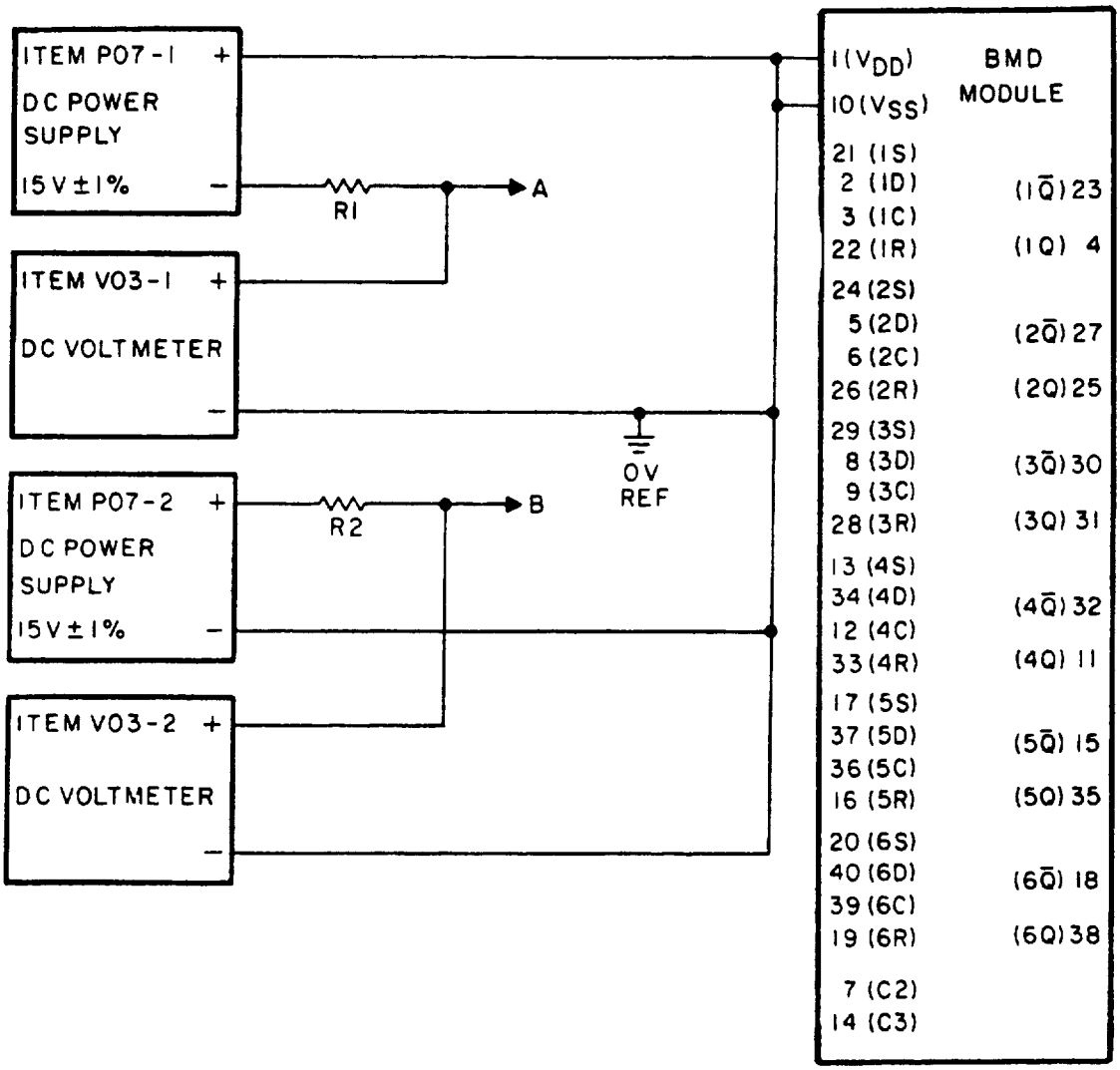
Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1S Input clamp diode voltage, positive	Open	21	0.4	1.5	V	7
C2	1D		2				
C3	1C		3				
C4	1R		22				
C5	2S		24				
C6	2D		5				
C7	2C		6				
C8	2R		26				
C9	3S		29				
C10	3D		8				
C11	3C		9				
C12	3R		28				
C13	4S		13				
C14	4D		34				
C15	4C		12				
C16	4R		33				
C17	5S		17				
C18	5D		37				
C19	5C		36				
C20	5R		16				

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TABLE V. Input clamp diode voltage (positive) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C21	6S Input clamp diode voltage, positive	Open	20	0.4	1.5	V	7
C22	6D	↓	40	↓	↓	↓	↓
C23	6C	↓	39	↓	↓	↓	↓
C24	6R	↓	19	↓	↓	↓	↓

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NOTES.
1. R1 = 9 09 kΩ ±1%, 1/8 W
2. R2 = 13 7 kΩ ±1%, 1/8 W
3. Since dc power supplies P07-1 and P07-2 and dc voltmeters V03-1 and V03-2 are not simultaneous, only one dc power supply P07 and one dc voltmeter V03, is required

FIGURE 7 Input clamp diode voltage (negative and positive) test circuit for tables IV and V

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TABLE VI. Static output voltage (low-level, I_{OL} = 0) test.

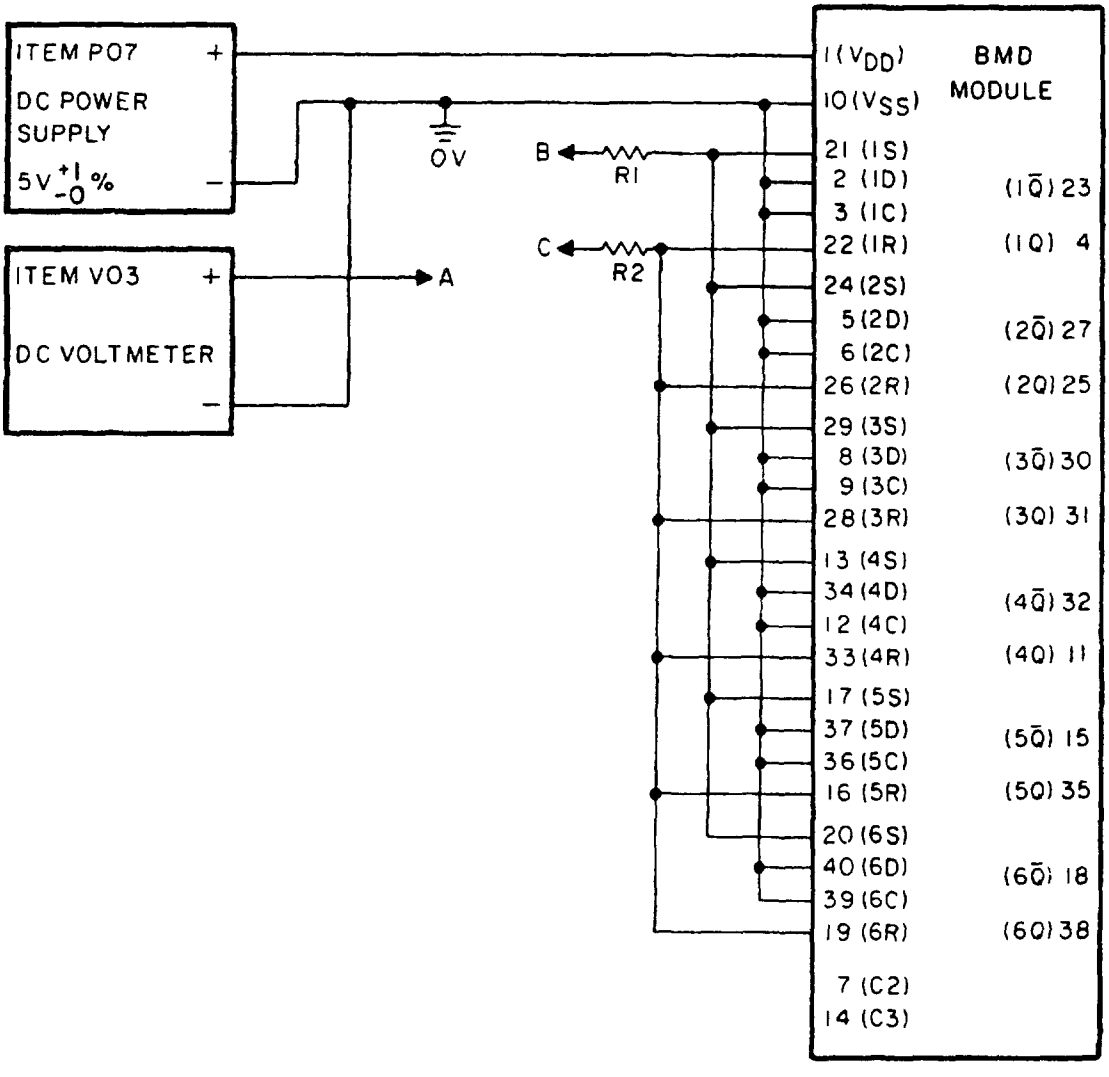
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1Q Static output voltage, low level, I_{OL} = 0	4	10	1	0 00	0.05	V	8
D2	2Q	25						
D3	3Q	31						
D4	4Q	11						
D5	5Q	35						
D6	6Q	38						
D7	1$\bar{Q}$	23	1	10				
D8	2$\bar{Q}$	27						
D9	3$\bar{Q}$	30						
D10	4$\bar{Q}$	32						
D11	5$\bar{Q}$	15						
D12	6$\bar{Q}$	18						

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TABLE VII. Static output voltage (high-level, I_{OH} = 0) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
E1	1Q Static output voltage, high level, I_{OH} = 0	23	10	1	4.95	5.05	V	8
E2	2Q	27	↓	↓	↓	↓	↓	↓
E3	3Q	30						
E4	4Q	32						
E5	5Q	15						
E6	6Q	18						
E7	$1\overline{Q}$	4	1	10				
E8	$2\overline{Q}$	25	↓	↓				
E9	$3\overline{Q}$	31						
E10	$4\overline{Q}$	11						
E11	$5\overline{Q}$	35						
E12	$6\overline{Q}$	38	↓	↓	↓	↓	↓	↓

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NOTE R1 = R2 = 1.5 kΩ ±1 , 1/4 W

FIGURE 8 Static output voltage (low-level, I_{OL} = 0 and high-level, I_{OH} = 0' test circuit for tables VI and VII

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TABLE VIII. Static output voltage (low-level, V_{DD} = 5 V) test.

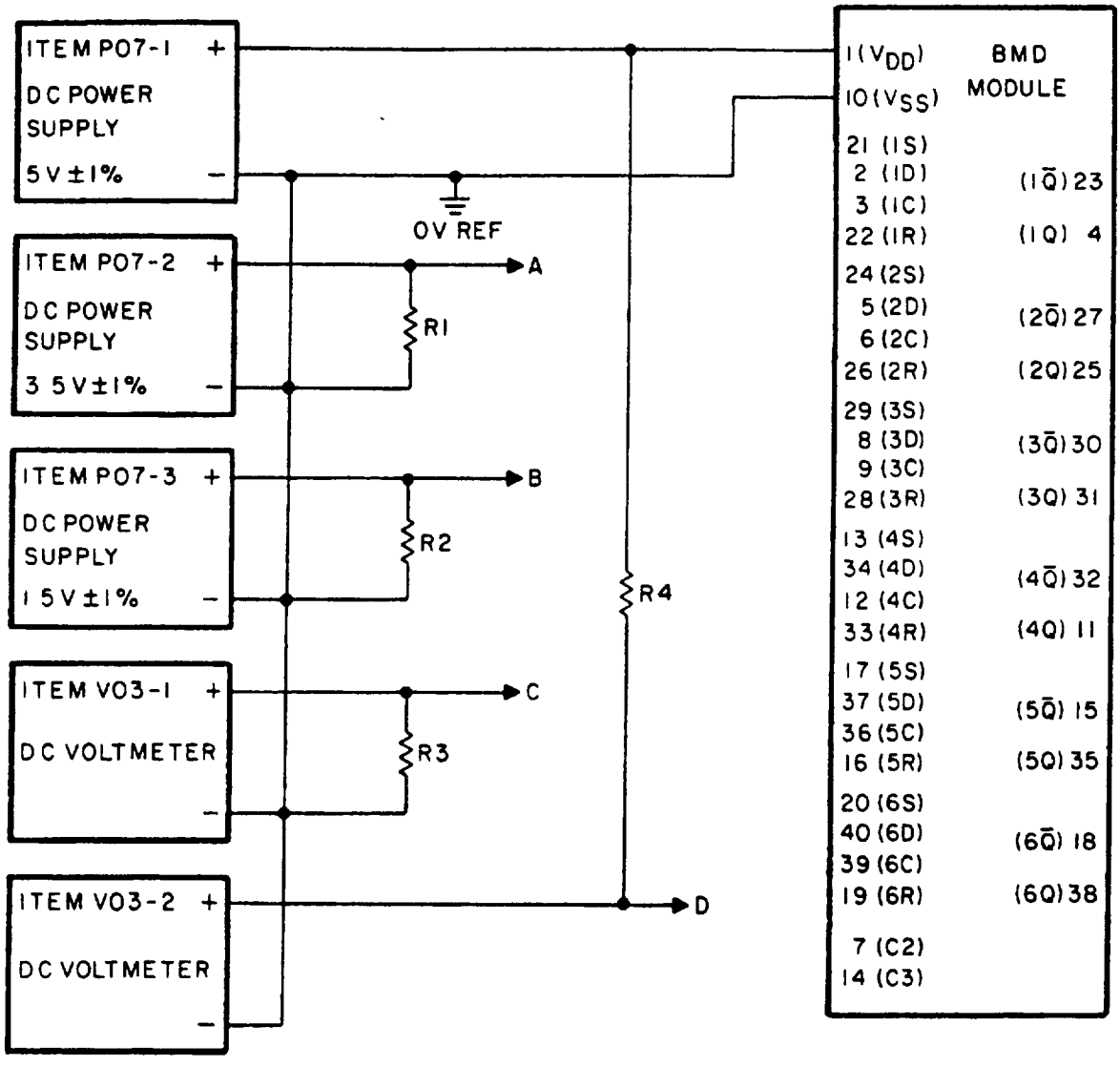
Test symbol	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
F1	1Q Static output voltage, low level, V_{DD} = 5 V	22	2,3,5,6,8,9,34,12,37, 36,40,39,21	Open	4	0.0	0.4	V	9
F2	2Q	26	2,3,5,6,8,9,34,12,37, 36,40,39,24		25				
F3	3Q	28	2,3,5,6,8,9,34,12,37, 36,40,39,29		31				
F4	4Q	33	2,3,5,6,8,9,34,12,37, 36,40,39,13		11				
F5	5Q	16	2,3,5,6,8,9,34,12,37, 36,40,39,17		35				
F6	6Q	19	2,3,5,6,8,9,34,12,37, 36,40,39,20		38				
F7	$1\overline{Q}$	21	2,3,5,6,8,9,34,12,37, 36,40,39,22		23				
F8	$2\overline{Q}$	24	2,3,5,6,8,9,34,12,37, 36,40,39,26		27				
F9	$3\overline{Q}$	29	2,3,5,6,8,9,34,12,37, 36,40,39,28		30				
F10	$4\overline{Q}$	13	2,3,5,6,8,9,34,12,37, 36,40,39,33		32				
F11	$5\overline{Q}$	17	2,3,5,6,8,9,34,12,37, 36,40,39,16		15				
F12	$6\overline{Q}$	20	2,3,5,6,8,9,34,12,37, 36,40,39,19		18				

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TABLE IX. Static output voltage (high-level, V_{DD} = 5 V) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	$1\overline{Q}$ Static output voltage, high level, V_{DD} = 5 V	22	2,3,5,6,8,9,34,12,37, 36,40,39,21	23	Open	4.6	5.0	V	9
G2	$2\overline{Q}$	26	2,3,5,6,8,9,34,12,37, 36,40,39,24	27					
G3	$3\overline{Q}$	28	2,3,5,6,8,9,34,12,37, 36,40,39,29	30					
G4	$4\overline{Q}$	33	2,3,5,6,8,9,34,12,37, 36,40,39,13	32					
G5	$5\overline{Q}$	16	2,3,5,6,8,9,34,12,37, 36,40,39,17	15					
G6	$6\overline{Q}$	19	2,3,5,6,8,9,34,12,37, 36,40,39,20	18					
G7	1Q	21	2,3,5,6,8,9,34,12,37, 36,40,39,22	4					
G8	2Q	24	2,3,5,6,8,9,34,12,37, 36,40,39,26	25					
G9	3Q	29	2,3,5,6,8,9,34,12,37, 36,40,39,28	31					
G10	4Q	13	2,3,5,6,8,9,34,12,37, 36,40,39,33	11					
G11	5Q	17	2,3,5,6,8,9,34,12,37, 36,40,39,16	35					
G12	6Q	20	2,3,5,6,8,9,34,12,37, 36,40,39,19	38					

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/8 W.
2. R3 = @ 0°C and 25°C 23.2 kΩ ±1%, 1/8 W.
   = @ 60°C 27.4 kΩ ±1%, 1/8 W.
   = @ -55°C 18.2 kΩ ±1%, 1/8 W.
   = @ 100°C 32 4 kΩ ±1%, 1/8 W.
3. R4 = @ 0°C and 25°C 9.09 kΩ ±1%, 1/8 W.
   = @ 60°C 10.5 kΩ ±1%, 1/8 W
   = @ -55°C 7.15 kΩ ±1%, 1/8 W
   = @ 100°C 12.7 kΩ ±1%, 1/8 W

FIGURE 9. Static output voltage (low-level, V_{DD} = 5 V and high-level, V_{DD} = 5 V) test circuit for tables VIII and IX.

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TABLE V. Static output voltage (low-level, V_{DD} = 15 V) test

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
H1	$1\overline{Q}$ Static output voltage, low level, V_{DD} = 15 V	22,26,28,33,16,19	21,24,29,13,17,20	23	0.0	1.5	V	10
H2	$2\overline{Q}$			27				
H3	$3\overline{Q}$			30				
H4	$4\overline{Q}$			32				
H5	$5\overline{Q}$			15				
H6	$6\overline{Q}$			18				
H7	1Q	21,24,29,13,17,20	22,26,28,33,16,19	4				
H8	2Q			25				
H9	3Q			31				
H10	4Q			11				
H11	5Q			35				
H12	6Q			38				

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TABLE XI. Static output voltage (high-level, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1Q Static output voltage, high level, V_{DD} = 15 V	22,26,28,33,16,19	21,24,29,13,17,20	4	13.5	15.0	V	10
J2	2Q			25				
J3	3Q			31				
J4	4Q			11				
J5	5Q			35				
J6	6Q			38				
J7	$1\bar{Q}$	21,24,29,13,17,20	22,26,28,33,16,19	23				
J8	$2\bar{Q}$			27				
J9	$3\bar{Q}$			30				
J10	$4\bar{Q}$			32				
J11	$5\bar{Q}$			15				
J12	$6\bar{Q}$			18				

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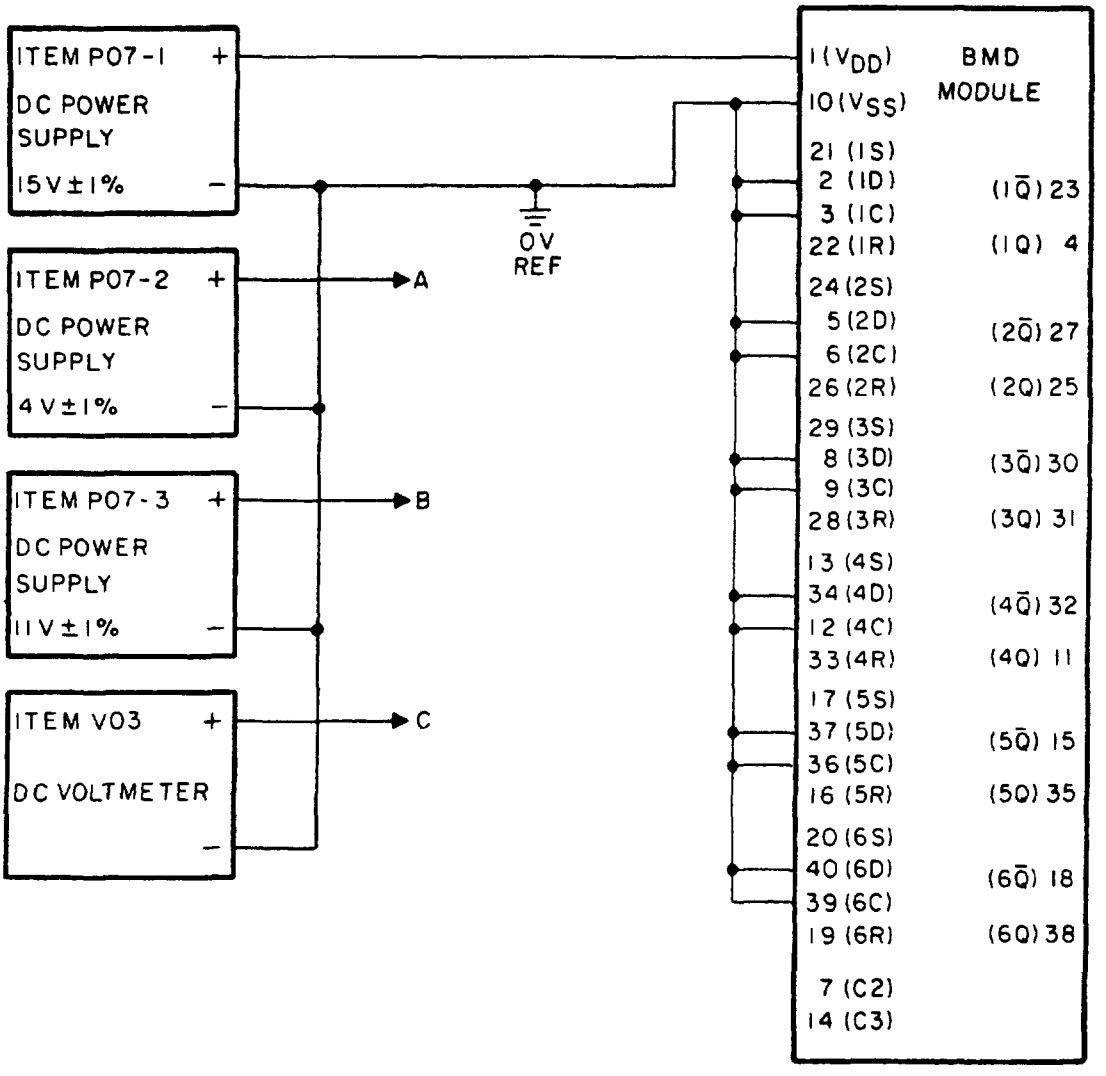


FIGURE 10 Static output voltage (low-level, V_{DD} = 15 V and high-level, V_{DD} = 15 V) test circuit for tables X and XI

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TABLE XII. Input current (low-level) test.

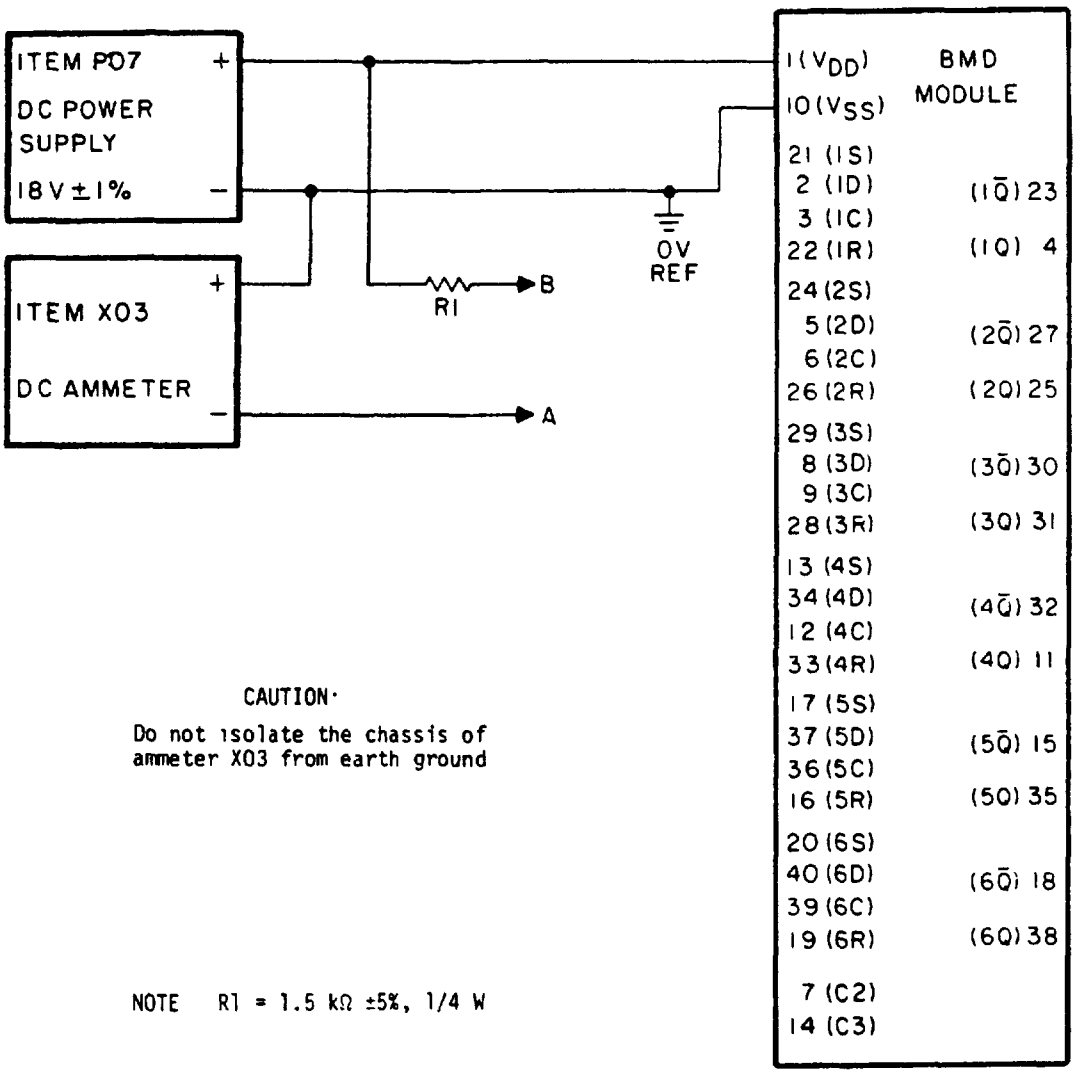
Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	1S Input current, low level	21	2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19	0	-200	nA	11
K2	1D	2	21,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
[illegible]	1C	3	21,2,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
K4	1R	22	21,2,3,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
K5	2S	24	21,2,3,22,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
K6	2D	5	21,2,3,22,24,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
K7	2C	6	21,2,3,22,24,5,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
K8	2R	26	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
K9	3S	29	21,2,3,22,24,5,6,26,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
K10	3D	8	21,2,3,22,24,5,6,26,29,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
K11	3C	9	21,2,3,22,24,5,6,26,29,8,28,13, 34,12,33,17,37,36,16,20,40,39,19				
K12	3R	28	21,2,3,22,24,5,6,26,29,8,9,13,34, 12,33,17,37,36,16,20,40,39,19				

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K13	4S Input current, low level	13	21,2,3,22,24,5,6,26,29,8,9,28,34 12,33,17,37,36,16,20,40,39,19	0	-200	nA	11
K14	4D	34	21,2,3,22,24,5,6,26,29,8,9,28,13, 12,33,17,37,36,16,20,40,39,19				
K15	4C	12	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,33,17,37,36,16,20,40,39,19				
K16	4R	33	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,17,37,36,16,20,40,39,19				
K17	5S	17	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,37,36,16,20,40,39,19				
K18	5D	37	21,2,3,22,24,5,6,26,29,8,9,28, 13,34,12,33,17,36,16,20,40,39,19				
K19	5C	36	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,16,20,40,39,19				
K20	5R	16	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,20,40,39,19				
K21	6S	20	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,40,39,19				
K22	6D	40	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,39,19				
K23	6C	39	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,19				
K24	6R	19	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39				

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CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground

NOTE R1 = 1.5 kΩ ±5%, 1/4 W

FIGURE 11. Input current (low-level) test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

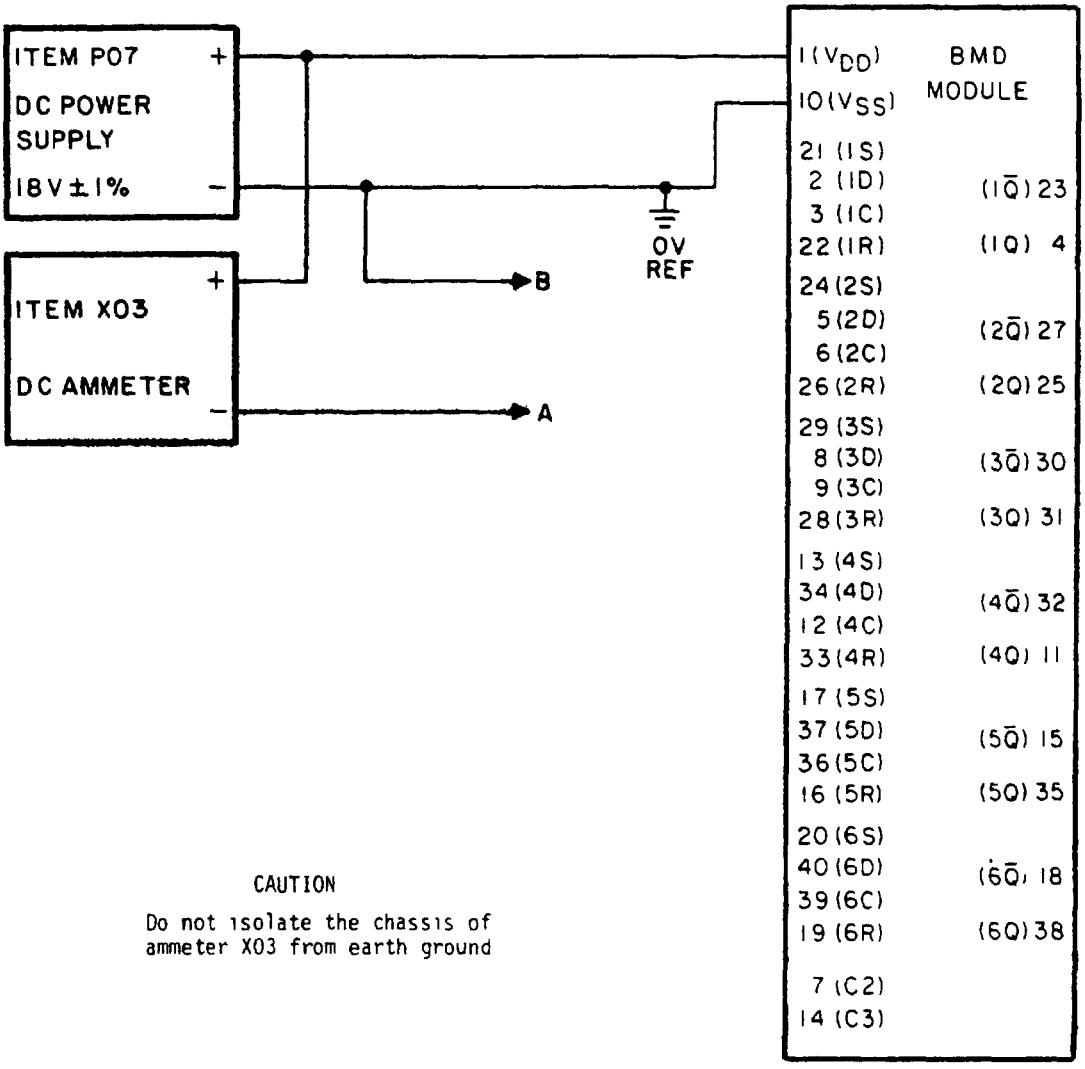
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
L1	1S Input current, high level	21	2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19	0	200	nA	12
L2	1D	2	21,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
L3	1C	3	21,2,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
L4	1R	22	21,2,3,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
L5	2S	24	21,2,3,22,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
L6	2D	5	21,2,3,22,24,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
L7	2C	6	21,2,3,22,24,5,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
L8	2R	26	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
L9	3S	29	21,2,3,22,24,5,6,26,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19				
L10	3D	8	21,2,3,22,24,5,6,26,29,9,28,13, 34,12,33,17,37,36,16,20,40,39,19				
L11	3C	9	21,2,3,22,24,5,6,26,29,8,28,13, 34,12,33,17,37,36,16,20,40,39,19				
L12	3R	28	21,2,3,22,24,5,6,26,29,8,9,13,34, 12,33,17,37,36,16,20,40,39,19				

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L13	4S Input current, high level	13	21,2,3,22,24,5,6,26,29,8,9,28,34 12,33,17,37,36,16,20,40,39,19	0	200	nA	12
L14	4D	34	21,2,3,22,24,5,6,26,29,8,9,28,13, 12,33,17,37,36,16,20,40,39,19				
L15	4C	12	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,33,17,37,36,16,20,40,39,19				
L16	4R	33	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,17,37,36,16,20,40,39,19				
L17	5S	17	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,37,36,16,20,40,39,19				
L18	5D	37	21,2,3,22,24,5,6,26,29,8,9,28, 13,34,12,33,17,36,16,20,40,39,19				
L19	5C	36	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,16,20,40,39,19				
L20	5R	16	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,20,40,39,19				
L21	6S	20	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,40,39,19				
L22	6D	40	21,2,3,22,24,5,6,26,29,8,9,28 13, 34,12,33,17,37,36,16,20,39,19				
L23	6C	39	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,19				
L24	6R	19	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,40,39				

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CAUTION

Do not isolate the chassis of ammeter X03 from earth ground

FIGURE 12. Input current (high-level) test circuit for table VIII

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TABLE XIV. Static power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Static power supply current, low level	21,24,29,13,17,20	22,2,3,26,5,6,28, 8,9,33,34,12,16, 37,36,19,40,39	0	60	µA	13
M2	Static power supply current, high level	2,3,22,5,6,26,8,9, 28,34,12,33,37,36, 16,40,39,19	21,24,29,13,17,20	0	60	µA	13

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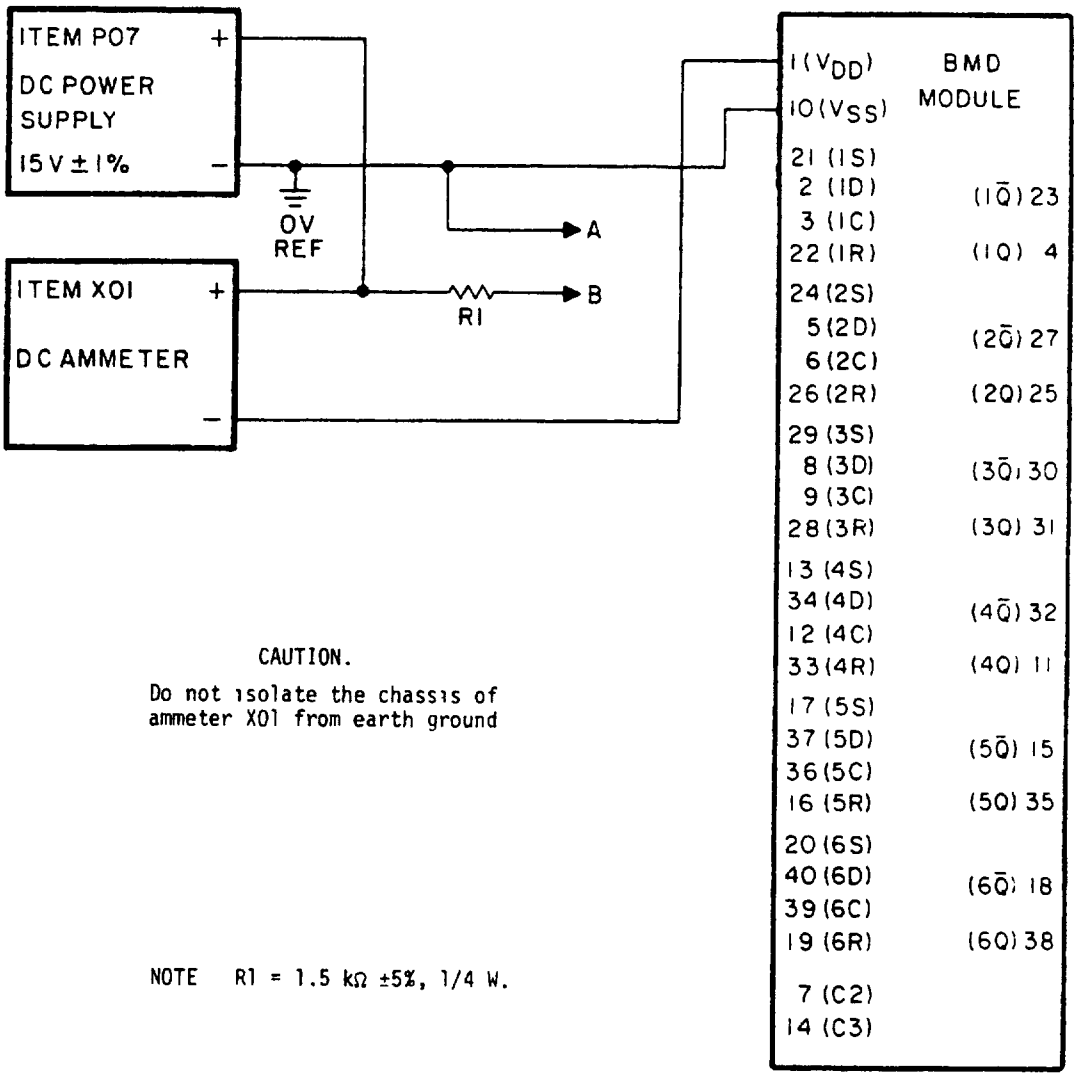


FIGURE 13. Static power supply current (low-level and high-level) test circuit for table XIV

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TABLE XV. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
N1	1Q Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	36	360	ns	14,15
N2	2Q	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
N3	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
N4	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
N5	5Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
N6	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
N7	1$\overline{Q}$	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
N8	2$\overline{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
N9	3$\overline{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
N10	4$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
N11	5$\overline{Q}$	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
N12	6$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N13	1Q Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	84	840	ns	14,15
N14	2Q	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
N15	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
N16	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
N17	5Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
N18	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
N19	1$\overline{Q}$	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
N20	2$\overline{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
N21	3$\overline{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
N22	4$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
N23	5$\overline{Q}$	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
N24	6$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XVI. Transition time (t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	$1\overline{Q}$ Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	20	200	ns	14,15
P2	$2\overline{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
P3	$3\overline{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
P4	$4\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
P5	$5\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
P6	$6\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
P7	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
P8	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
P9	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
P10	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
P11	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
P12	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

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TABLE XVI Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P13	$1\bar{0}$ Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	68	680	ns	14,15
P14	$2\bar{0}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
P15	$3\bar{0}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
P16	$4\bar{0}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
P17	$5\bar{0}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
P18	$6\bar{0}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
P19	10	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
P20	20	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
P21	30	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
P22	40	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
P23	50	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
P24	60	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

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TABLE XVII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1Q	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	35	350	ns	14,16
Q2	2Q		21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
Q3	3Q		21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
Q4	4Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
Q5	5Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
Q6	6Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
Q7	$1\bar{Q}$		21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
Q8	$2\bar{Q}$		21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
Q9	$3\bar{Q}$		21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
Q10	$4\bar{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
Q11	$5\bar{Q}$		21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
Q12	$6\bar{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XVII Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
Q13	1Q Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	55	550	ns	14,16
Q14	2Q	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
Q15	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
Q16	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
Q17	5Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
Q18	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
Q19	1$\bar{Q}$	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
Q20	2$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
Q21	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
Q22	4$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
Q23	5$\bar{Q}$	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
Q24	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:07Z
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TABLE XVIII. Propagation delay time (t_{PHL}) test.

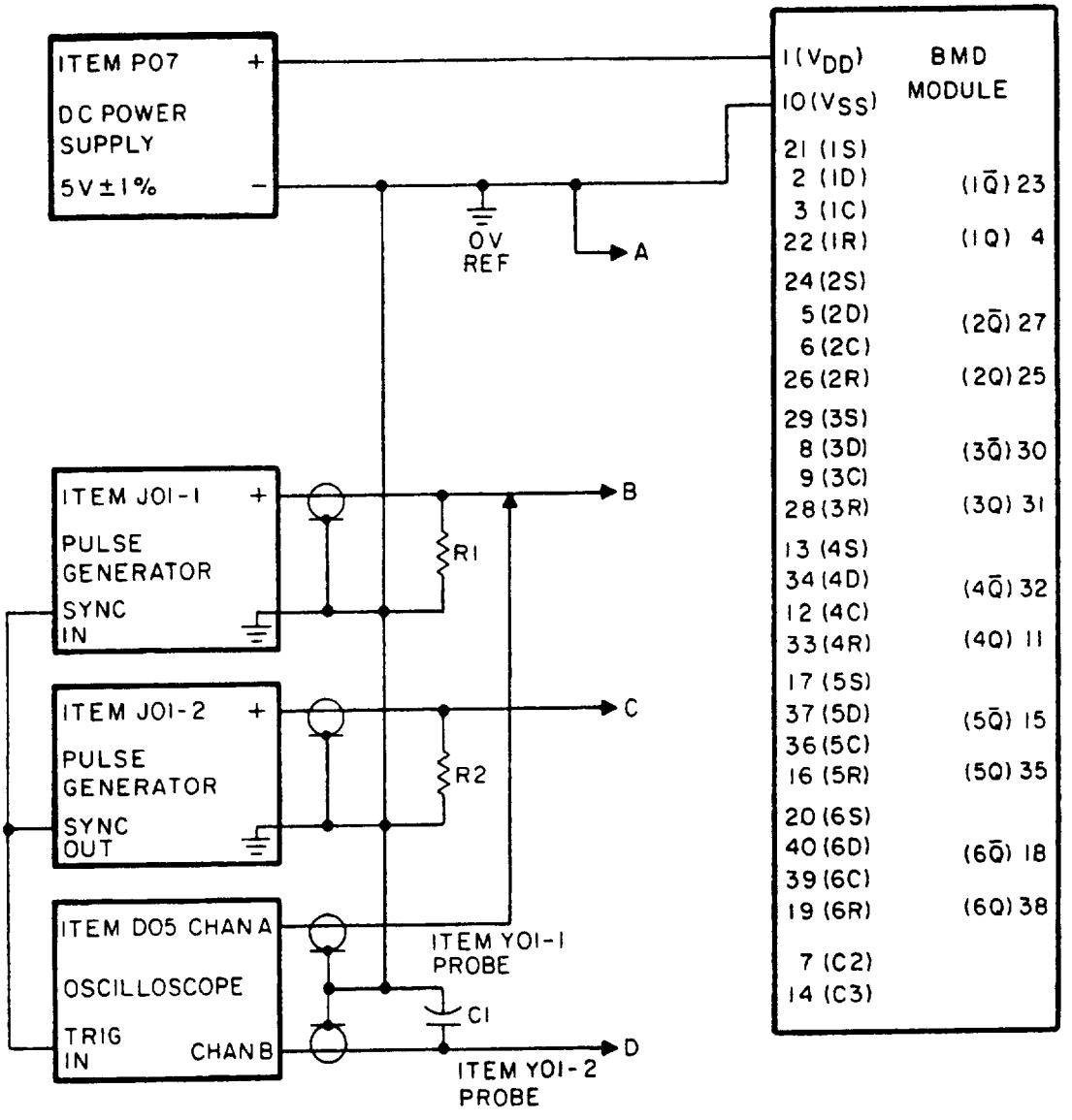
Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R1	$1\bar{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	35	350	ns	14,16
R2	$2\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
R3	$3\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
R4	$4\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
R5	$5\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
R6	$6\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
R7	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
R8	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
R9	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
R10	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
R11	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
R12	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
R13	1$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	55	550	ns	14,16
R14	2$\overline{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
R15	3$\overline{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
R16	4$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
R17	5$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
R18	6$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
R19	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
R20	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
R21	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
R22	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
R23	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
R24	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

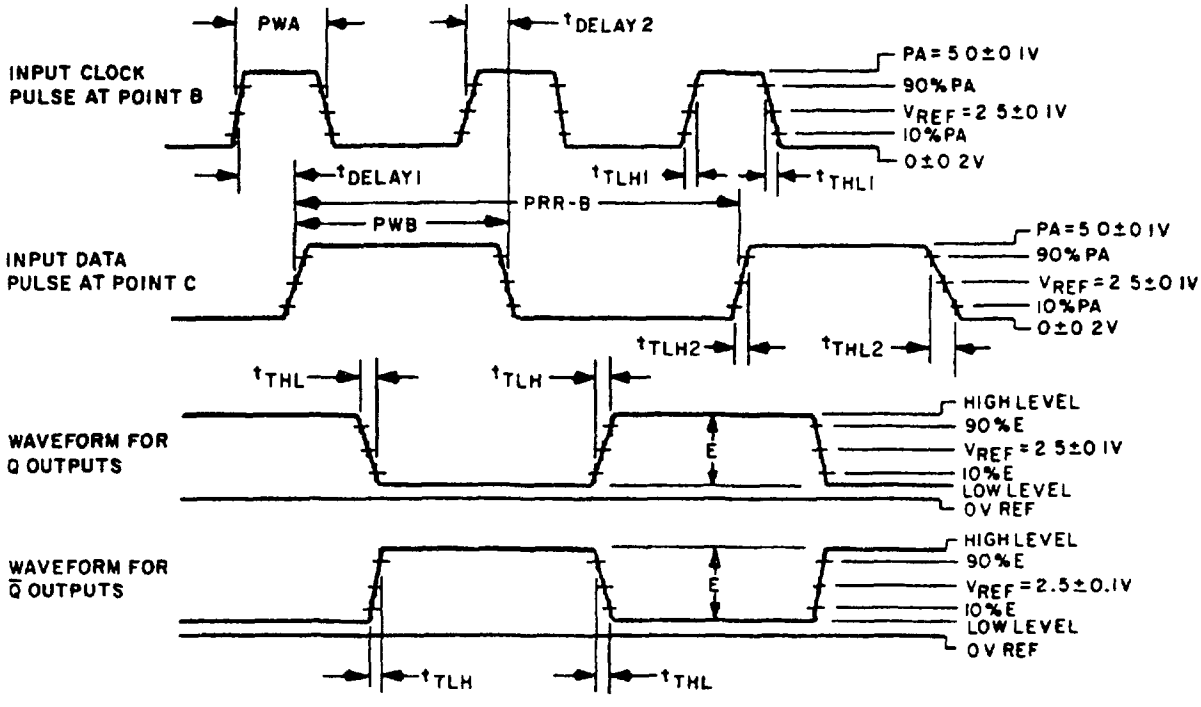
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NOTES
1. R1 = R2 = 51.1Ω ±5%, 1/2 W
2. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N12, P1-P12, Q1-Q12 and R1-R12.
3. C1 = 150 ±15 pF (including probe and parasitic capacitance) for test conditions N13-N24, P13-P24, Q13-Q24 and R13-R24.

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV, XVI, XVII and XVIII

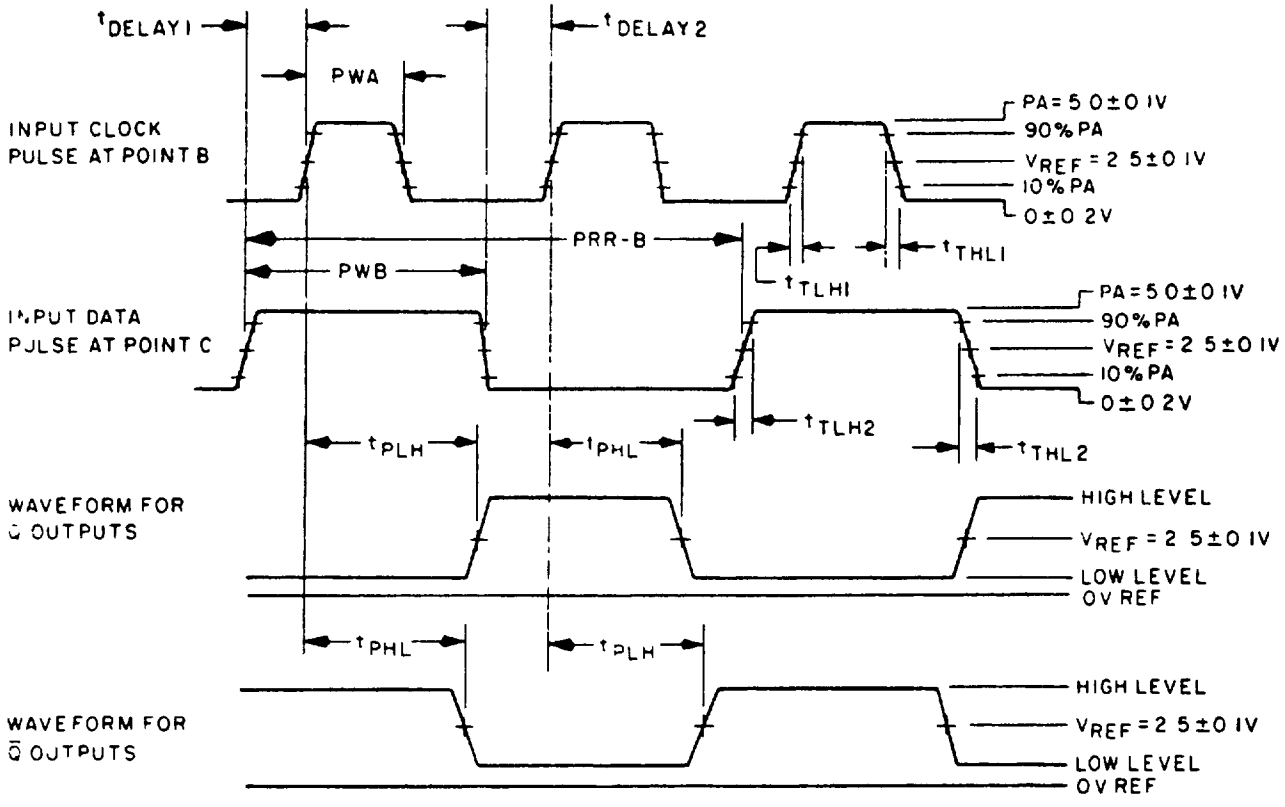
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NOTES.
1. Clock input pulse parameters
   PWA = 1.0 ±0.1 µs
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   Put item J01-1 in double pulse mode
2. Data input pulse parameters
   PWB = 2.0 ±0.2 µs.
   PRR-B = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns
3. $t_{DELAY1} = t_{DELAY2} = 100$ +10, -0 ns
4. Before making measurements, verify that the output pulse high level is ≥ 4.6 V and that the low level is ≤ 0.4 V
5. The lead length of electrical conductors between terminals, resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between lead circuit and output pins shall be less than 1.00 inch (25.4 mm)
6. All voltages are referenced to ground

FIGURE 15. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XV and XVI

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NOTES
1 Clock input pulse parameters
PWA = 1 0 ±0 1 µs
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns
Put item J01-1 in double pulse mode
2 Data input pulse parameters
PWB = 2 0 ±0.2 µs
PRR-B = 4 0 ±0.4 µs.
$t_{TLH2} = t_{THL2}$ = 10 ±2 ns
3 $t_{DELAY1} = t_{DELAY2}$ = 100 +10, -0 ns.
4 Before making measurements, verify that the output pulse high level is ≥ 4.6 V and that the low level is ≤ 0 4 V.
5 The lead length of electrical conductors between terminals, resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between lead circuit and output pins shall be less than 1 00 inch (25.4 mm)
6 All voltages are referenced to ground

FIGURE 16 Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVII and XVIII

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TABLE XIX. Propagation delay time (t_{PLH}) test

Test condition	Parameters	Connect					Initial limits 25°±5°C		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	F to pin	Min	Max		
S1	1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	24,26,29,28,13,33,17,16,20,19	21	22	B	4	35	350	ns	17,18
S2	2Q	21,22,29,28,13,33,17,16,20,19	24	26		25				
S3	3Q	21,22,24,26,13,33,17,16,20,19	29	28		31				
S4	4Q	21,22,24,26,29,28,17,16,20,19	13	33		11				
S5	5Q	21,22,24,26,29,28,13,33,20,19	17	16		35				
S6	6Q	21,22,24,26,29,28,13,33,17,16	20	19		38				
S7	$1\bar{Q}$	24,26,29,28,13,33,17,16,20,19	21	22	C	23	45	450		
S8	$2\bar{Q}$	21,22,29,28,13,33,17,16,20,19	24	26		27				
S9	$3\bar{Q}$	21,22,24,26,13,33,17,16,20,19	29	28		30				
S10	$4\bar{Q}$	21,22,24,26,29,28,17,16,20,19	13	33		32				
S11	$5\bar{Q}$	21,22,24,26,29,28,13,33,20,19	17	16		15				
S12	$6\bar{Q}$	21,22,24,26,29,28,13,33,17,16	20	19		18				
S13	1Q Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	24,26,29,28,13,33,17,16,20,19	21	22	B	4	55	550		
S14	2Q	21,22,29,28,13,33,17,16,20,19	24	26		25				
S15	3Q	21,22,24,26,13,33,17,16,20,19	29	28		31				
S16	4Q	21,22,24,26,29,28,17,16,20,19	13	33		11				
S17	5Q	21,22,24,26,29,28,13,33,20,19	17	16		35				
S18	6Q	21,22,24,26,29,28,13,33,17,16	20	19		38				
S19	$1\bar{Q}$	24,26,29,28,13,33,17,16,20,19	21	22	C	4	65	650		
S20	$2\bar{Q}$	21,22,29,28,13,33,17,16,20,19	24	26		27				
S21	$3\bar{Q}$	21,22,24,26,13,33,17,16,20,19	29	28		30				
S22	$4\bar{Q}$	21,22,24,26,29,28,17,16,20,19	13	33		32				
S23	$5\bar{Q}$	21,22,24,26,29,28 13,33,20,19	17	16		15				
S24	$6\bar{Q}$	21,22,24,26,29,28 13,33,17,16	20	19		18				

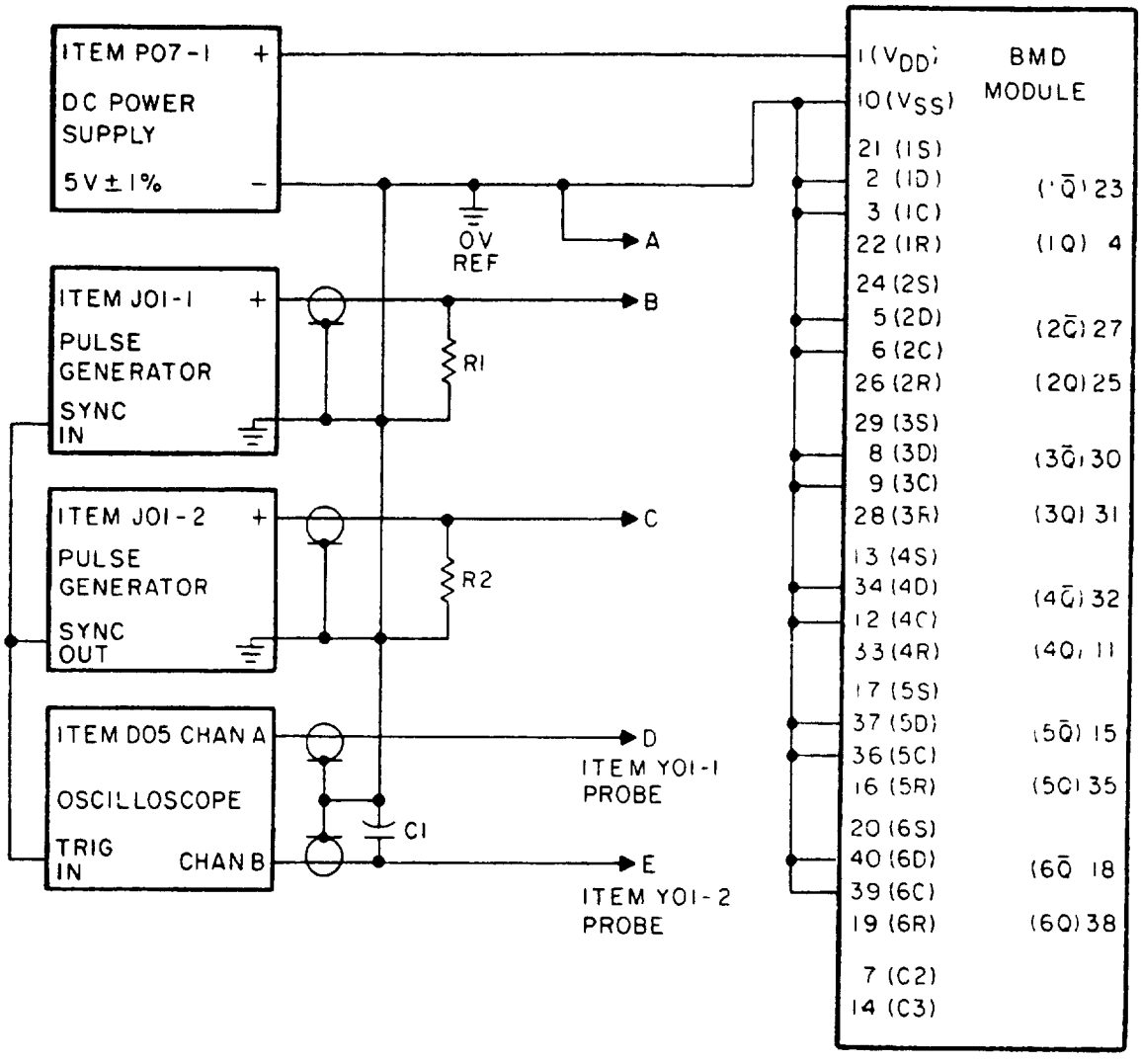
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TABLE XX. Propagation delay time (t_{PHL}) test.

Test condition	Parameters		Connect					Initial limits 25°+5°C		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	$1\bar{Q}$	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	24,26,29,28,13,33,17,16,20,19	21	22	B	23	35	350	ns	17,18
T2	$2\bar{Q}$		21,22,29,28,13,33,17,16,20,19	24	26		27				
T3	$3\bar{Q}$		21,22,24,26,13,33,17,16,20,19	29	28		30				
T4	$4\bar{Q}$		21,22,24,26,29,28,17,16,20,19	13	33		32				
T5	$5\bar{Q}$		21,22,24,26,29,28,13,33,20,19	17	16		15				
T6	$6\bar{Q}$		21,22,24,26,29,28,13,33,17,16	20	19		18				
T7	1Q		24,26,29,28,13,33,17,16,20,19	21	22	C	4	45	450		
T8	2Q		21,22,29,28,13,33,17,16,20,19	24	26		25				
T9	3Q		21,22,24,26,13,33,17,16,20,19	29	28		31				
T10	4Q		21,22,24,26,29,28,17,16,20,19	13	33		11				
T11	5Q		21,22,24,26,29,28,13,33,20,19	17	16		35				
T12	6Q		21,22,24,26,29,28,13,33,17,16	20	19		38				
T13	$1\bar{Q}$	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	24,26,29,28,13,33,17,16,20,19	21	22	B	23	55	550		
T14	$2\bar{Q}$		21,22,29,28,13,33,17,16,20,19	24	26		27				
T15	$3\bar{Q}$		21,22,24,26,13,33,17,16,20,19	29	28		30				
T16	$4\bar{Q}$		21,22,24,26,29,28,17,16,20,19	13	33		32				
T17	$5\bar{Q}$		21,22,24,26,29,28,13,33,20,19	17	16		15				
T18	$6\bar{Q}$		21,22,24,26,29,28,13,33,17,16	20	19		18				
T19	1Q		24,26,29,28,13,33,17,16,20,19	21	22	C	4	65	650		
T20	2Q		21,22,29,28,13,33,17,16,20,19	24	26		25				
T21	3Q		21,22,24,26,13,33,17,16,20,19	29	28		31				
T22	4Q		21,22,24,26,29,28,17,16,20,19	13	33		11				
T23	5Q		21,22,24,26,29,28,13,33,20,19	17	16		35				
T24	6Q		21,22,24,26,29,28,13,33,17,16	20	19		38				

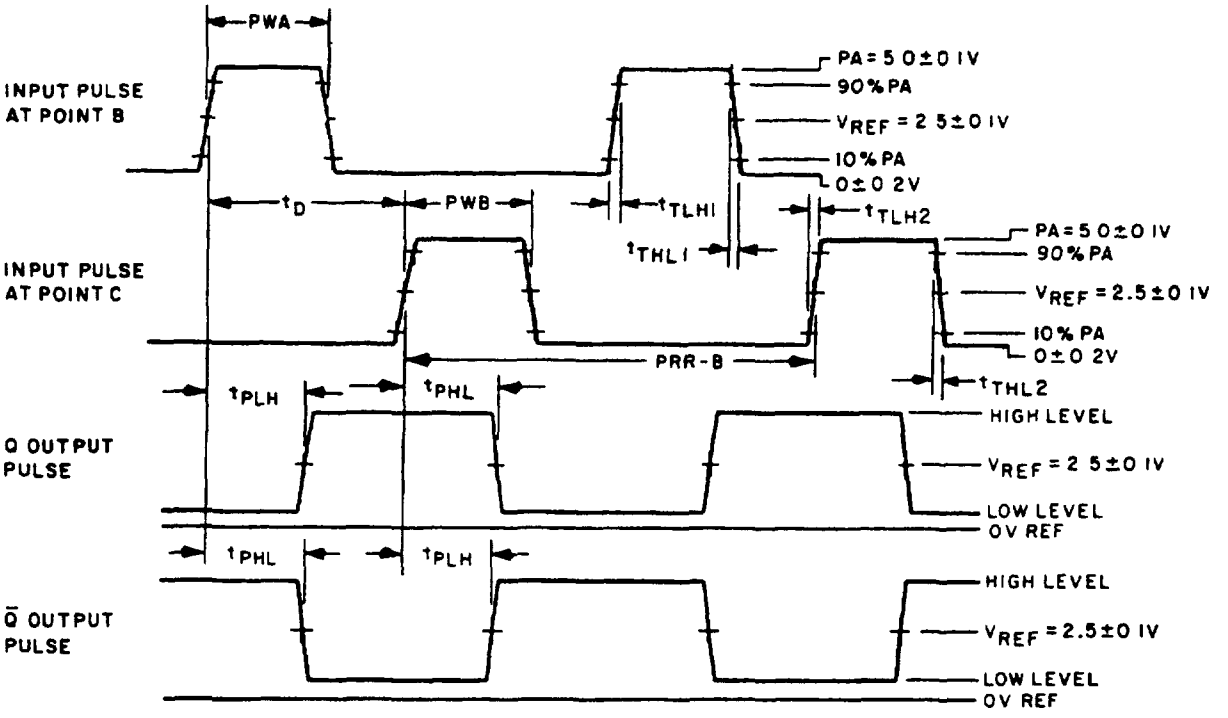
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NOTES
1. R1 = R2 = 51 Ω ±5%, 1/2 W
2. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions S1-S12, T1-T12
3. C1 = 150 ±15 pF (including probe and parasitic capacitance) for test conditions S13-S24, T13-T24

FIGURE 17 Propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIX and XX

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NOTES
1. Item J01-1 input pulse parameters
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 2 0 ±0 1 µs
2. Item J01-2 input pulse parameters
   PWB = 2.0 ±0 2 µs
   PRR-B = 4 0 ±0 4 µs
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns
3. $t_D = 2.0 \pm 0.2$ µs
4. Before making the measurements, verify that the output pulse high level is ≥ 4.6 V and that the low level is ≤ 0 4 V
5. The lead length of electrical conductors between terminator resistors R1 and R2 and output pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1 00 inch (25 4 mm).
6. All voltages are referenced to ground

FIGURE 18 Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIX and XX

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TABLE XXI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
U1	1Q Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	18	180	ns	19,20
U2	2Q	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
U3	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
U4	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
U5	5Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
U6	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
U7	1$\bar{Q}$	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
U8	2$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
U9	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
U10	4$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
U11	5$\bar{Q}$	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
U12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XXII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	$1\bar{Q}$ Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	10	100	ns	19,20
V2	$2\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
V3	$3\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
V4	$4\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
V5	$5\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
V6	$6\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
V7	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
V8	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
V9	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
V10	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
V11	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
V12	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

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TABLE XXIII. Propagation delay time (t_{PLH}) test.

Test condition		Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W1	1Q	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	15	150	ns	19,21
W2	2Q		21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
W3	3Q		21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
W4	4Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
W5	5Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
W6	6Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
W7	$1\overline{Q}$		21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
W8	$2\overline{Q}$		21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
W9	$3\overline{Q}$		21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
W10	$4\overline{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
W11	$5\overline{Q}$		21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
W12	$6\overline{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

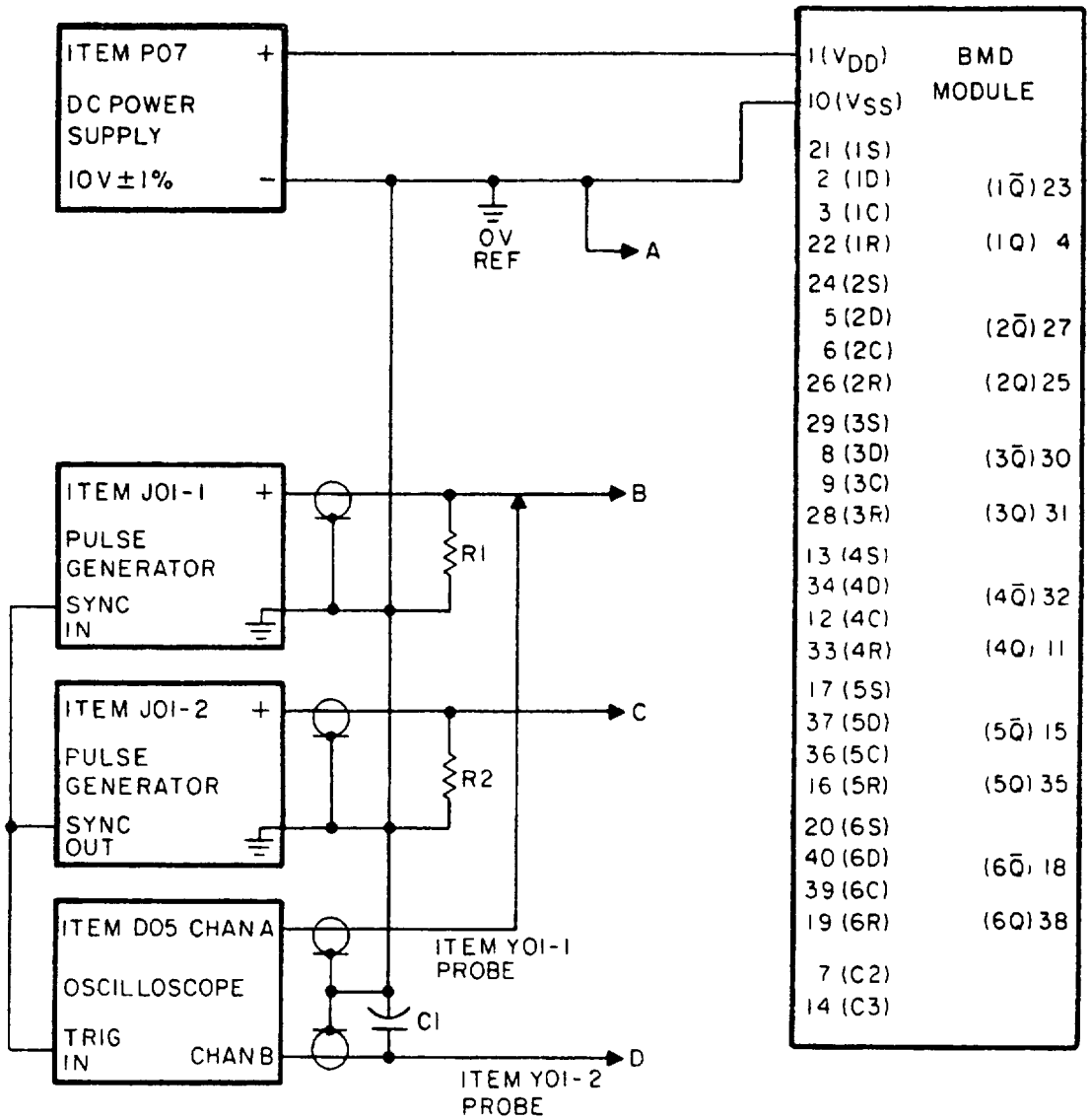
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TABLE XXIV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
X1	$1\bar{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	15	150	ns	19,21
X2	$2\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
X3	$3\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
X4	$4\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
X5	$5\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
X6	$6\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
X7	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
X8	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
X9	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
X10	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
X11	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
X12	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

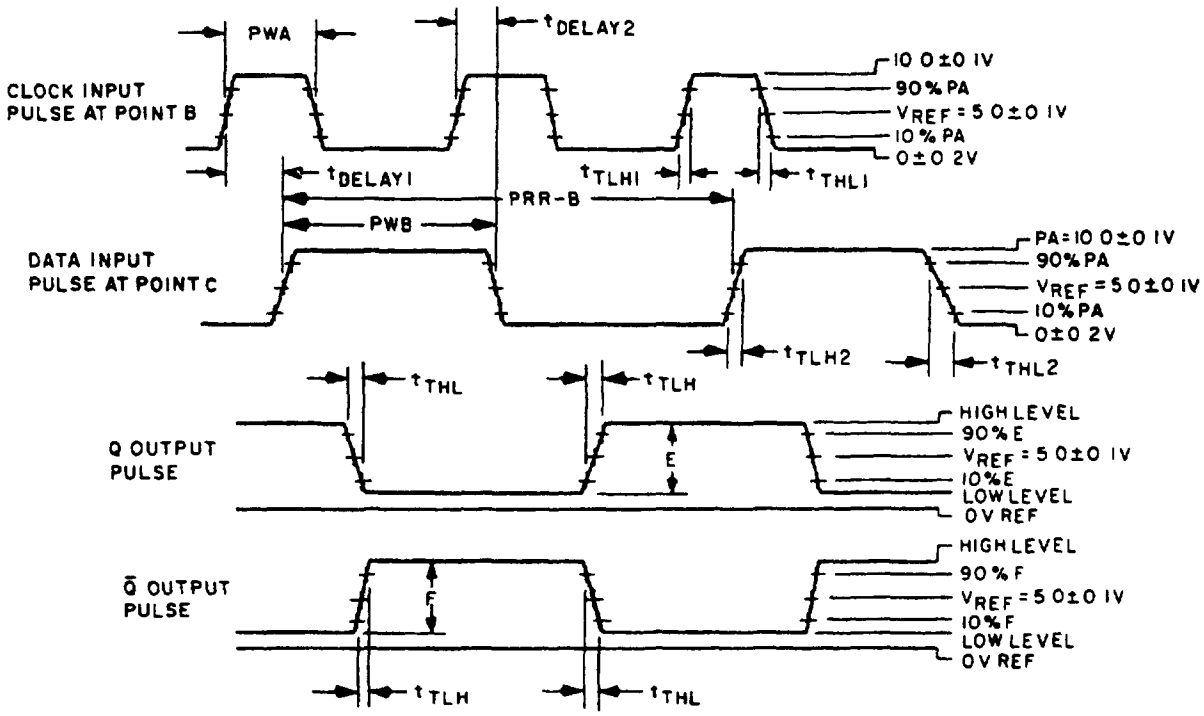
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NOTES
1. R1 = R2 = 51.1Ω ±5%, 2 W
2. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions U1-U12, V1-V12, W1-W12, and X1-X12

FIGURE 19. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXI through XXIV

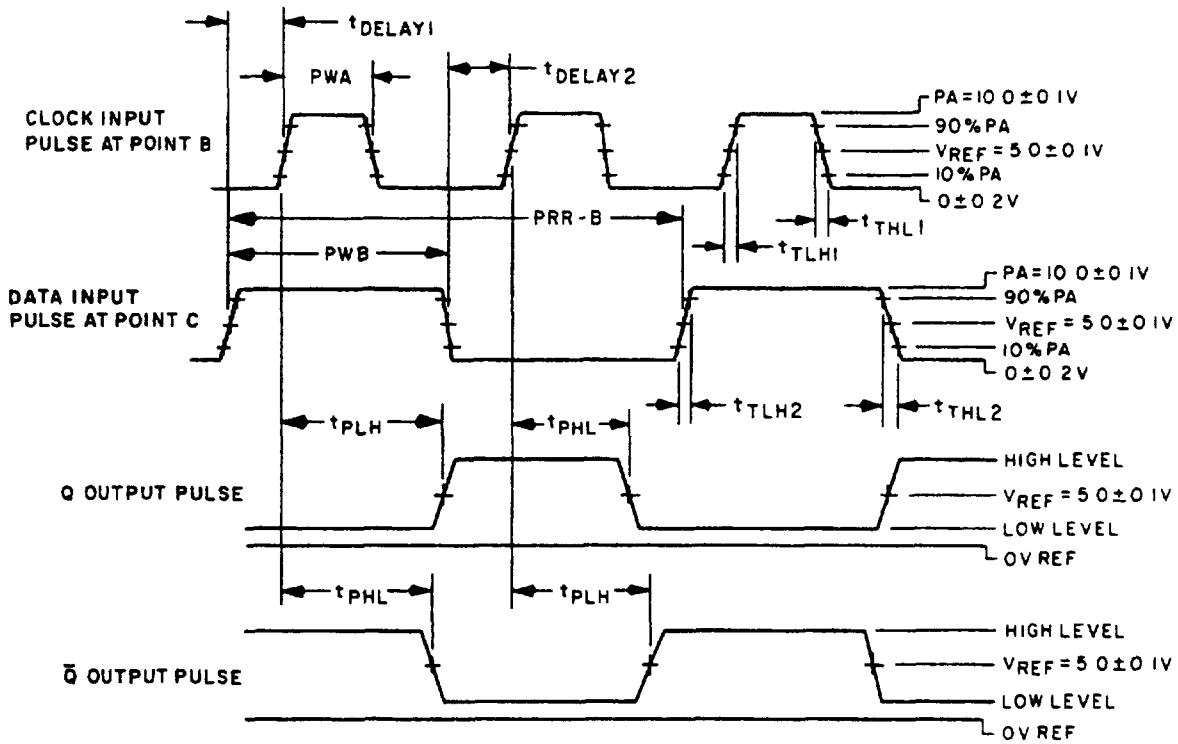
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NOTES:
1. Item J01-1 input pulse parameters
   PWA = 1.0 ±0.1 µs
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns
   Put item J01-1 in double pulse mode
2. Item J01-2 input pulse parameters
   PWB = 2.0 ±0.2 µs.
   PRR-B = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns
3. $t_{DELAY1} = t_{DELAY2}$ = 100 +10, -0 ns
4. Before making the measurements, verify that the output pulse high level is ≥ 9.5 V and that the low level is ≤ 0.5 V.
5. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 20. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXI and XXII.

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NOTES.
1. Item J01-1 input pulse parameters
   PWA = 1 0 ±0.1 µs
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns
   Put item J01-1 in double pulse mode
2. Item J01-2 input pulse parameters
   PWB = 2.0 ±0.2 µs
   PRR-B = 4 0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns
3. $t_{DELAY1} = t_{DELAY2}$ = 100 +10, -0 ns
4. Before making the measurements, verify that the output pulse high level is ≥ 9 5 V and that the low level is ≤ 0.5 V.
5. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1 00 inch (25.4 mm)
6. All voltages are referenced to ground

FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIII and XXIV

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TABLE XXV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	F to pin	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
Y1	$1\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	24,26,29,28,13,33,17,16,20,19	21	22	B	23	15	150	ns	22,23
Y2	$2\overline{Q}$	21,22,29,28,13,33,17,16,20,19	24	26		27				
Y3	$3\overline{Q}$	21,22,24,26,13,33,17,16,20,19	29	28		30				
Y4	$4\overline{Q}$	21,22,24,26,29,28,17,16,20,19	13	33		32				
Y5	$5\overline{Q}$	21,22,24,26,29,28,13,33,20,29	17	16		15				
Y6	$6\overline{Q}$	21,22,24,26,29,28,13,33,17,16	20	19		18				
Y7	1Q	24,26,29,28,13,33,17,16,20,19	21	22	C	4	20	200		
Y8	2Q	21,22,29,28,13,33,17,16,29,19	24	26		25				
Y9	3Q	21,22,24,26,13,33,17,16,20,19	29	28		31				
Y10	4Q	21,22,24,26,29,28,17,16,20,19	13	33		11				
Y11	5Q	21,22,24,26,29,28,13,33,20,19	17	16		35				
Y12	6Q	21,22,24,26,29,28,13,33,17,16	20	19		38				

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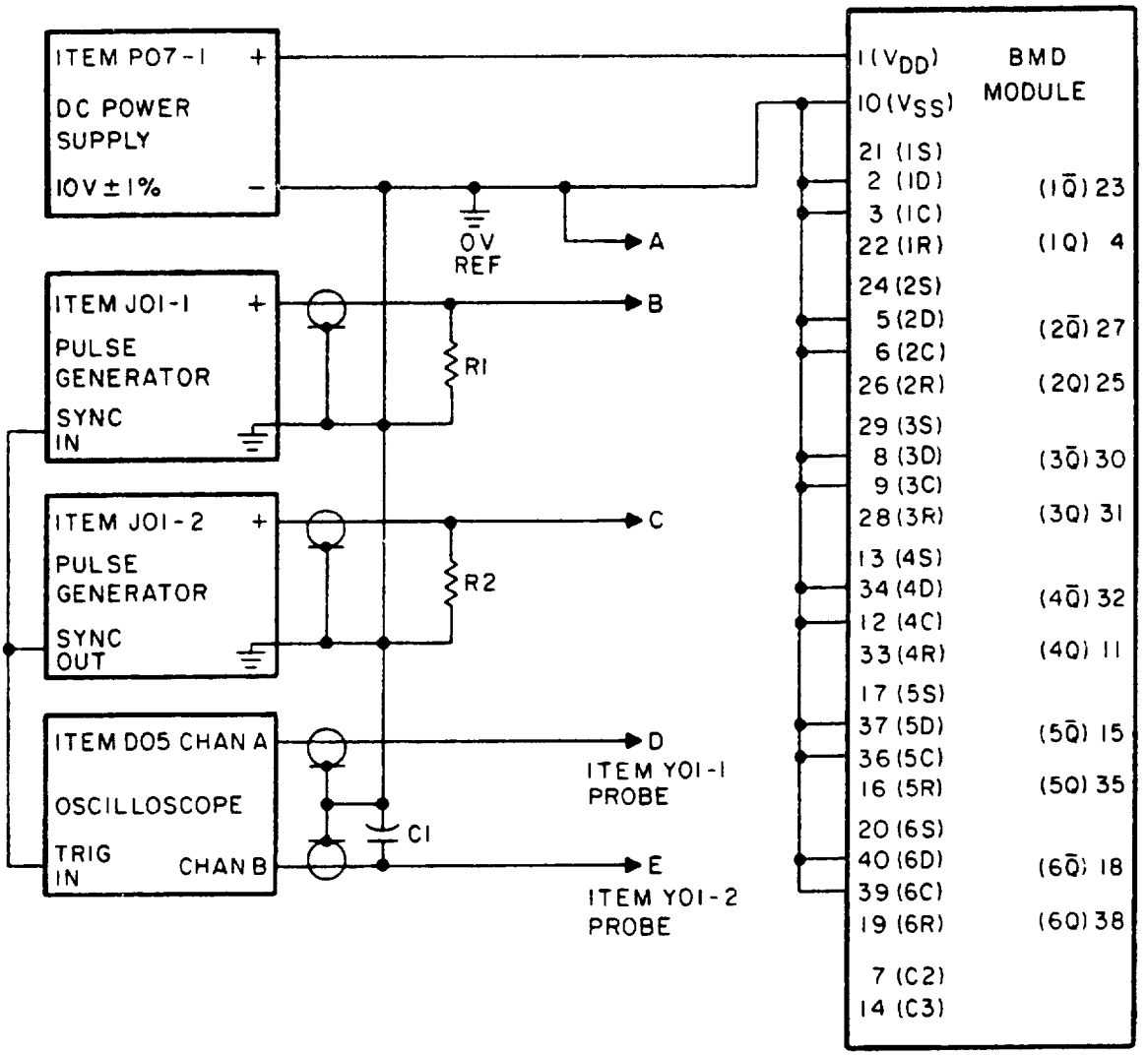
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TABLE XXVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
Z1	1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	24,26,29,28,13,33,17,16,20,19	21	22	B	4	15	150	ns	22,23
Z2	2Q	21,22,29,28,13,33,17,16,20,19	24	26		25				
Z3	3Q	21,22,24,26,13,33,17,16,20,19	29	28		31				
Z4	4Q	21,22,24,26,29,28,17,16,20,19	13	33		11				
Z5	5Q	21,22,24,26,29,28,13,33,20,19	17	16		35				
Z6	6Q	21,22,24,26,29,28,13,33,17,16	20	19		38				
Z7	1$\overline{Q}$	24,26,29,28,13,33,17,16,20,19	21	22	C	23	20	200		
Z8	2$\overline{Q}$	21,22,29,28,13,33,17,16,29,19	24	26		27				
Z9	3$\overline{Q}$	21,22,24,26,13,33,17,16,20,19	29	28		30				
Z10	4$\overline{Q}$	21,22,24,26,29,28,17,16,20,19	13	33		32				
Z11	5$\overline{Q}$	21,22,24,26,29,28,13,33,20,19	17	16		15				
Z12	6$\overline{Q}$	21,22,24,26,29,28,13,33,17,16	20	19		18				

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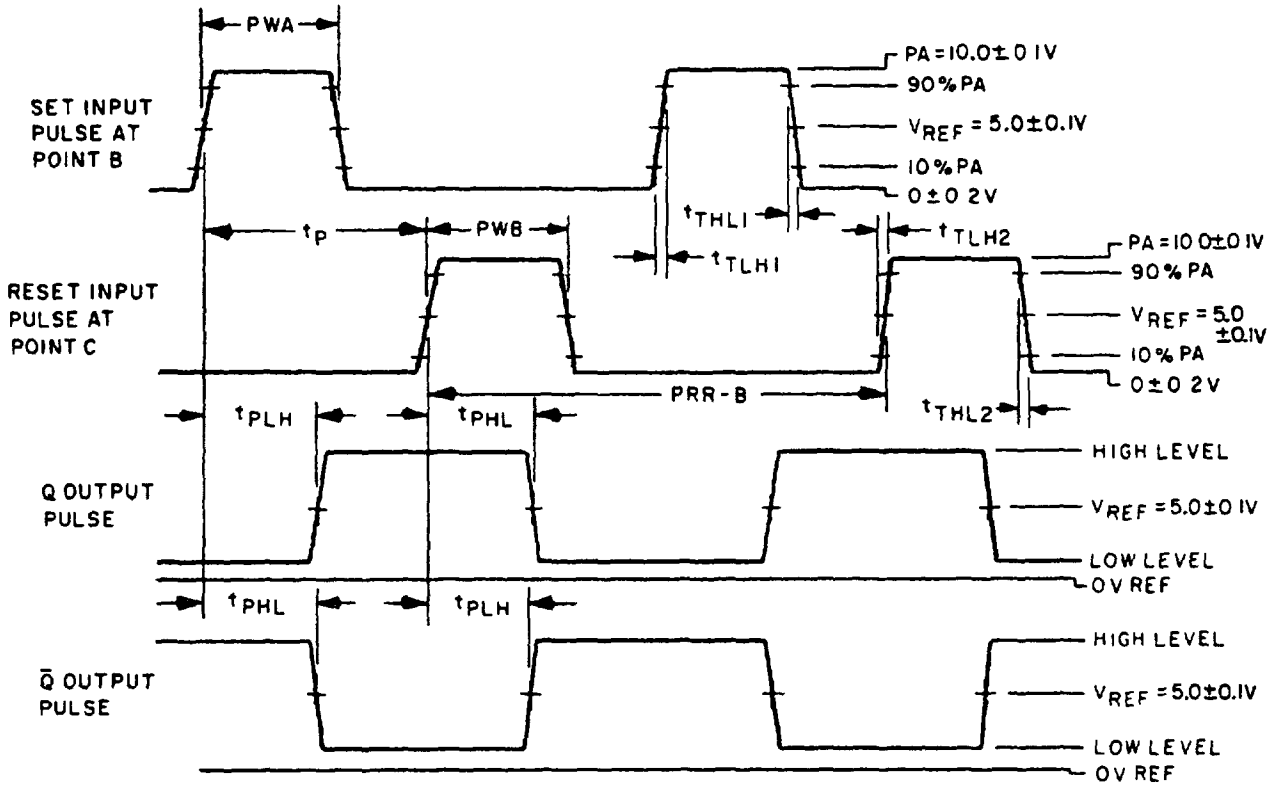
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NOTES
1 R1 = R2 = 51 Ω ±5%, 2 W
2 C1 = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 22 Propgation delay time (t_{PHL} and t_{PLH}) test circuit for tables XXV and XXVI.

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NOTES:
1. Item J01-1 input pulse parameters
   PWA = 2.0 ±0.2 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRR-B = 4.0 ±0.4 µs
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. t_D = 2.0 ±0.2 µs.
4. Before making the measurements, verify that the output pulse high level is ≥ 9.5 V and that the low level is ≤ 0.5 V.
5. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 23. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XXV and XXVI.

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TABLE XXVII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
AA1	1Q Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	13	130	ns	24,25
AA2	2Q	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
AA3	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
AA4	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
AA5	5Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
AA6	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
AA7	1$\bar{Q}$	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
AA8	2$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
AA9	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
AA10	4$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
AA11	5$\bar{Q}$	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
AA12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XXVIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
bB1	$1\overline{Q}$ Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	8	80	ns	24,25
bB2	$2\overline{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
BB3	$3\overline{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
BB4	$4\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
BB5	$5\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
bB6	$6\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
BB7	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
bB8	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
BB9	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
BB10	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
BB11	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
BB12	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

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TABLE XXIX. Propagation delay time (t_{PLH}) test.

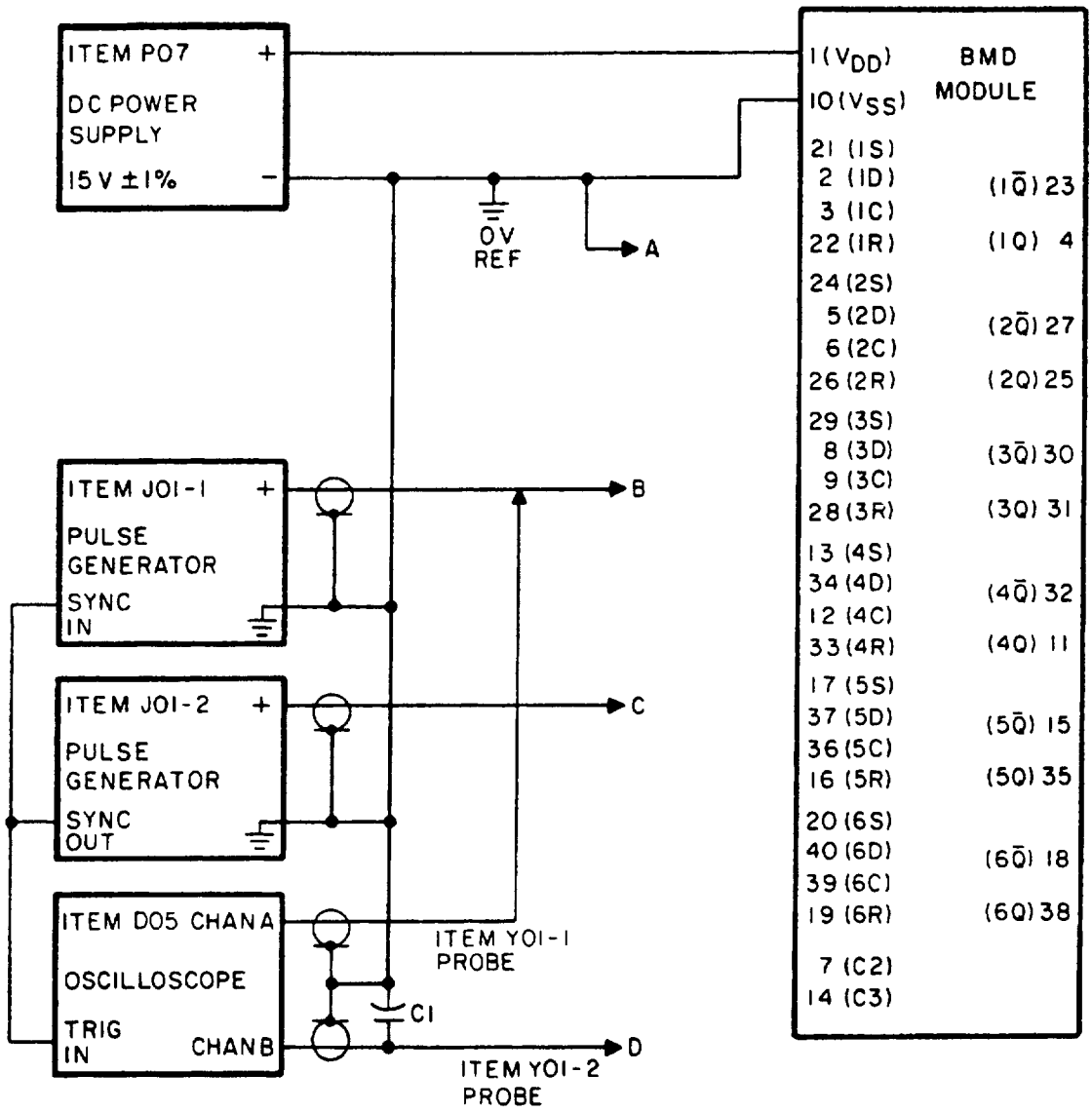
Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
CC1	1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	10	100	ns	24,26
CC2	2Q	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	25				
CC3	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
CC4	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
CC5	5Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	35				
CC6	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				
CC7	1$\bar{Q}$	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	23				
CC8	2$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	27				
CC9	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
CC10	4$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
CC11	5$\bar{Q}$	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	15				
CC12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XXX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
DD1	$1\bar{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	23	10	100	ns	24,26
DD2	$2\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	6	5	27				
DD3	$3\bar{Q}$	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	30				
DD4	$4\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	32				
DD5	$5\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	36	37	15				
DD6	$6\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				
DD7	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	3	2	4				
DD8	2Q	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
DD9	3Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	9	8	31				
DD10	4Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	12	34	11				
DD11	5Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	36	37	35				
DD12	6Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	38				

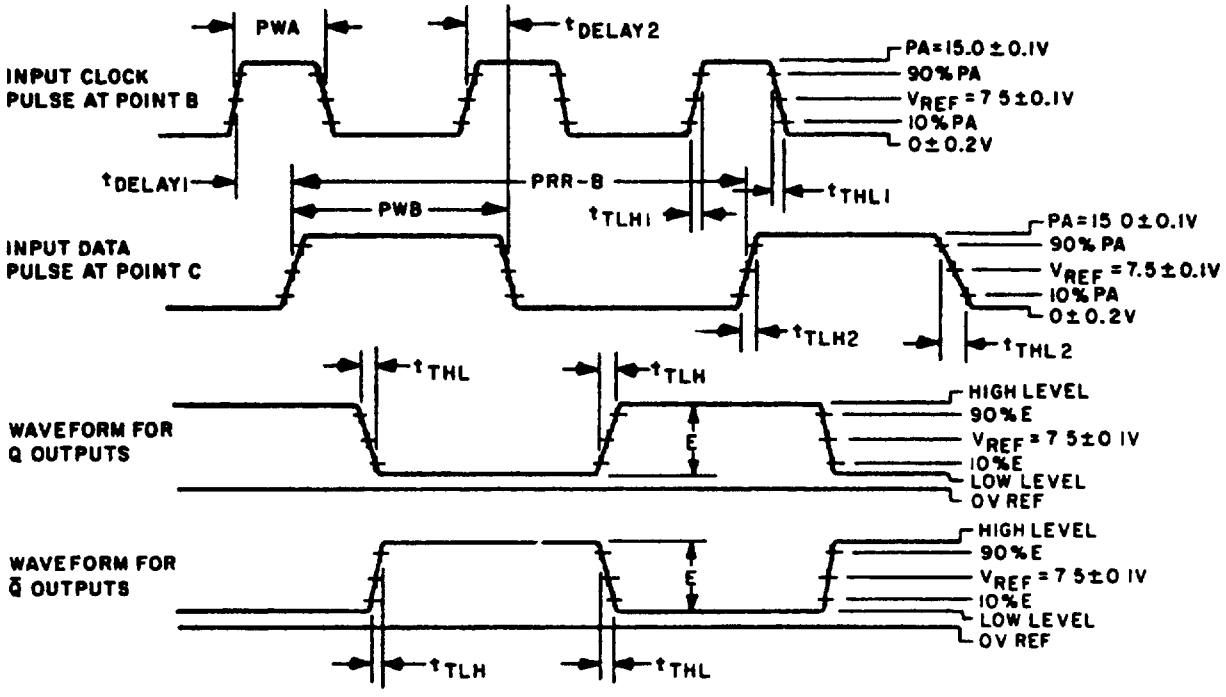
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NOTE C1 = 50 ±5 pF (including probe and parasitic capacitance)

FIGURE 24 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXVII through XXX.

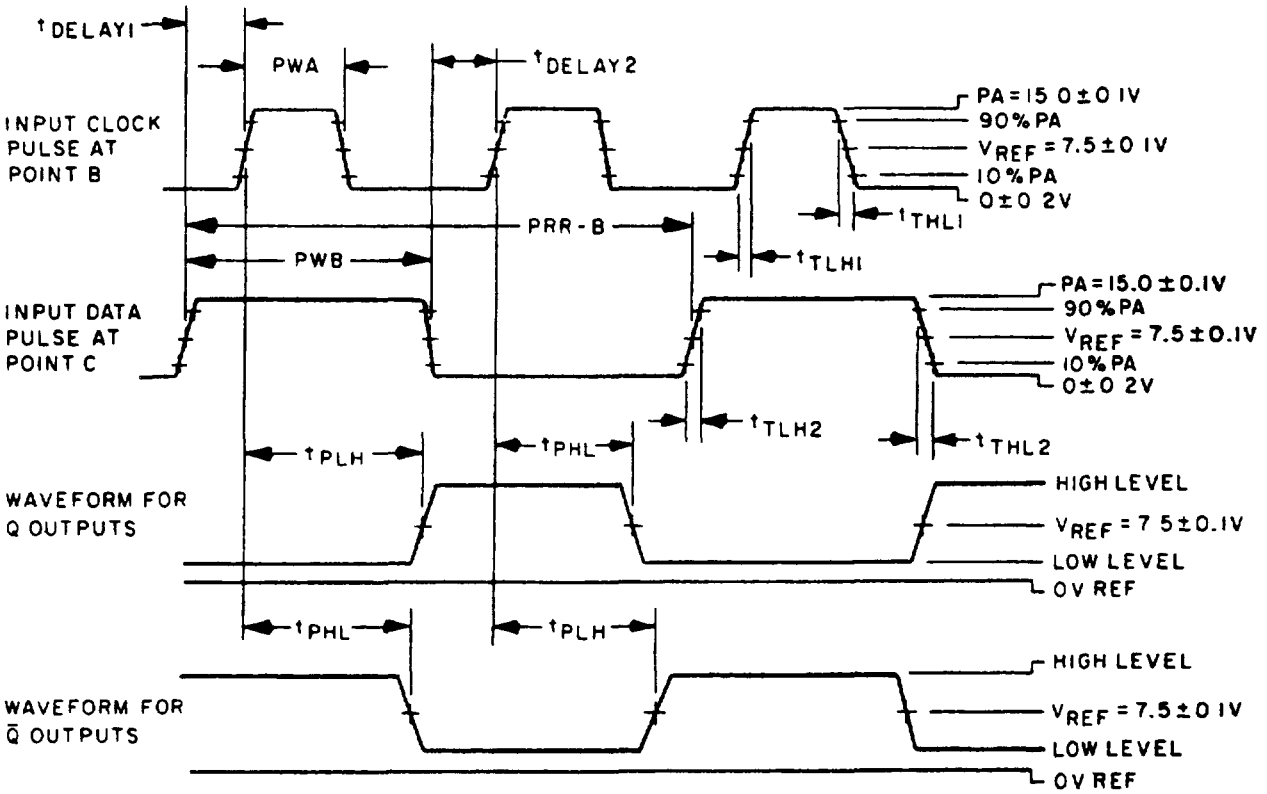
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NOTES:
1. Clock input pulse parameters.
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in double pulse mode.
2. Data input pulse parameters.
   PWB = 2.0 ±0.2 µs.
   PRR-B = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 100 +10, -0 ns.
4. Before making measurements, verify that the output pulse high level is ≥ 13.5 V and that the low level is ≤ 1.5 V.
5. The lead length of electrical conductors between input pins, probes Y01-1 and Y01-2 to input and output pins and between lead circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 25. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXVII and XXVIII.

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NOTES
1. Clock input pulse parameters
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   Put item J01-1 in double pulse mode
2. Data input pulse parameters
   PWB = 2.0 ±0.2 µs.
   PRR-B = 4.0 ±0.4 µs
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
3. $t_{DELAY1} = t_{DELAY2} = 100$ +10, -0 ns.
4. Before making measurements, verify that the output pulse high level is $\geq$ 13.5 V and that the low level is $\leq$ 1.5 V.
5. The lead length of electrical conductors between terminals, resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between lead circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground

FIGURE 26. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIX and XXX.

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TABLE XXXI. Propagation delay time (t_{PLH}) test.

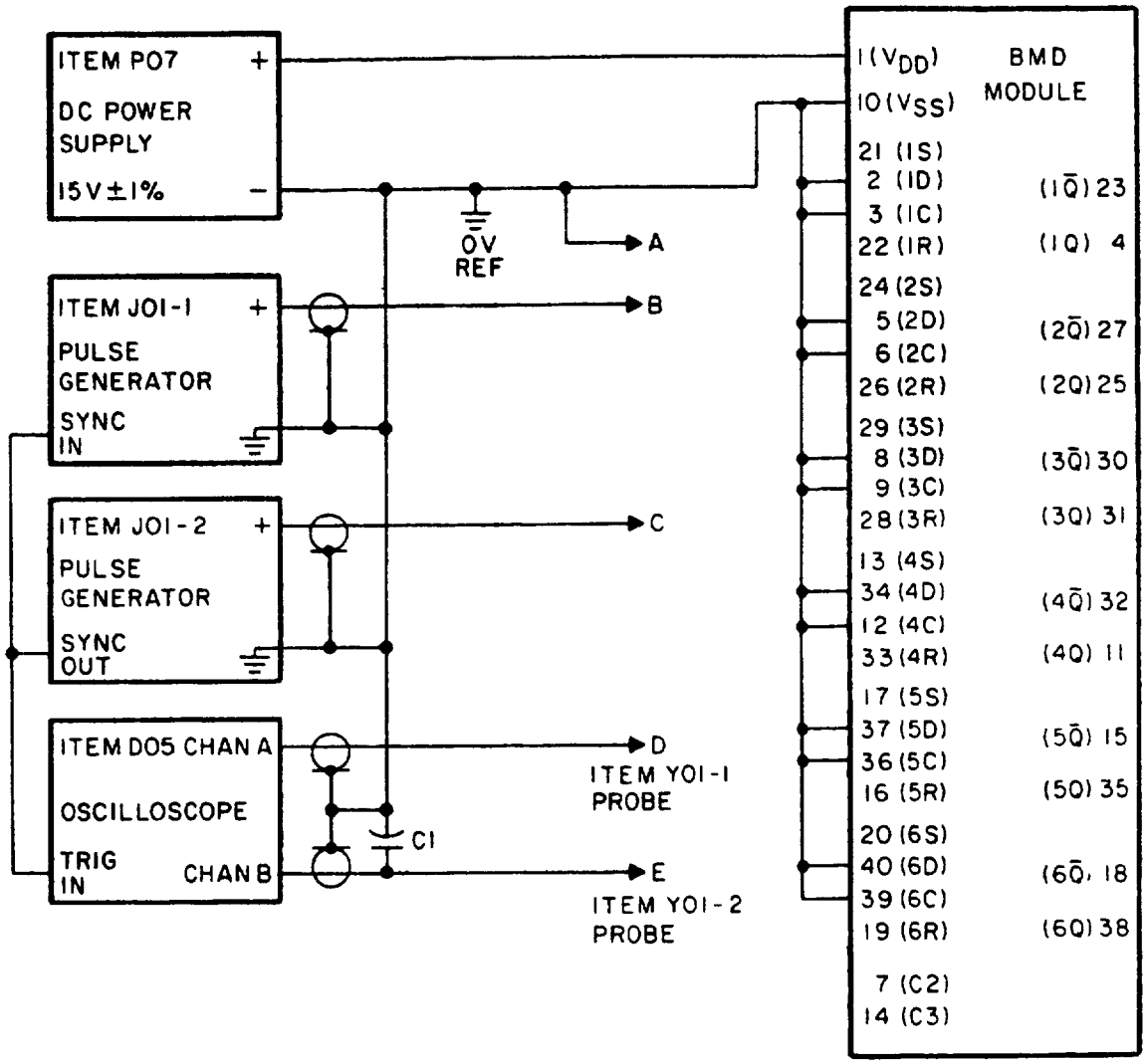
Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
EE1	1Q Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	24,26,29,28,13,33,17,16,20,19	21	22	B	4	10	100	ns	27,28
EE2	2Q	21,22,29,28,13,33,17,16,20,19	24	26		25				
EE3	3Q	21,22,24,26,13,33,17,16,20,19	29	28		31				
EE4	4Q	21,22,24,26,29,28,17,16,20,19	13	33		11				
EE5	5Q	21,22,242,6,29,28,13,33,20,19	17	16		35				
EE6	6Q	21,22,24,26,29,28,13,33,17,16	20	19		38				
EE7	1$\bar{Q}$	24,26,29,28,13,33,17,16,20,19	21	22	C	23	15	150		
EE8	2$\bar{Q}$	21,22,29,28,13,33,17,16,29,19	24	26		27				
EE9	3$\bar{Q}$	21,22,24,26,13,33,17,16,20,19	29	28		30				
EE10	4$\bar{Q}$	21,22,24,26,29,28,17,16,20,19	13	33		32				
EE11	5$\bar{Q}$	21,22,24,26,29,28,13,33,20,19	17	16		15				
EE12	6$\bar{Q}$	21,22,24,26,29,28,13,33,17,16	20	19		18				

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TABLE XXXII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
FF1	$1\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	24,26,29,28,13,33,17,16,20,19	21	22	B	23	10	100	ns	27,28
FF2	$2\overline{Q}$	21,22,29,28,13,33,17,16,20,19	24	26		27				
FF3	$3\overline{Q}$	21,22,24,26,13,33,17,16,20,19	29	28		30				
FF4	$4\overline{Q}$	21,22,24,26,29,28,17,16,20,19	13	33		32				
FF5	$5\overline{Q}$	21,22,242,6,29,28,13,33,20,19	17	16		15				
FF6	$6\overline{Q}$	21,22,24,26,29,28,13,33,17,16	20	19		18				
FF7	1Q	24,26,29,28,13,33,17,16,20,19	21	22	C	4	15	150		
FF8	2Q	21,22,29,28,13,33,17,16,29,19	24	26		25				
FF9	3Q	21,22,24,26,13,33,17,16,20,19	29	28		31				
FF10	4Q	21,22,24,26,29,28,17,16,20,19	13	33		11				
FF11	5Q	21,22,24,26,29,28,13,33,20,19	17	16		35				
FF12	6Q	21,22,24,26,29,28,13,33,17,16	20	19		38				

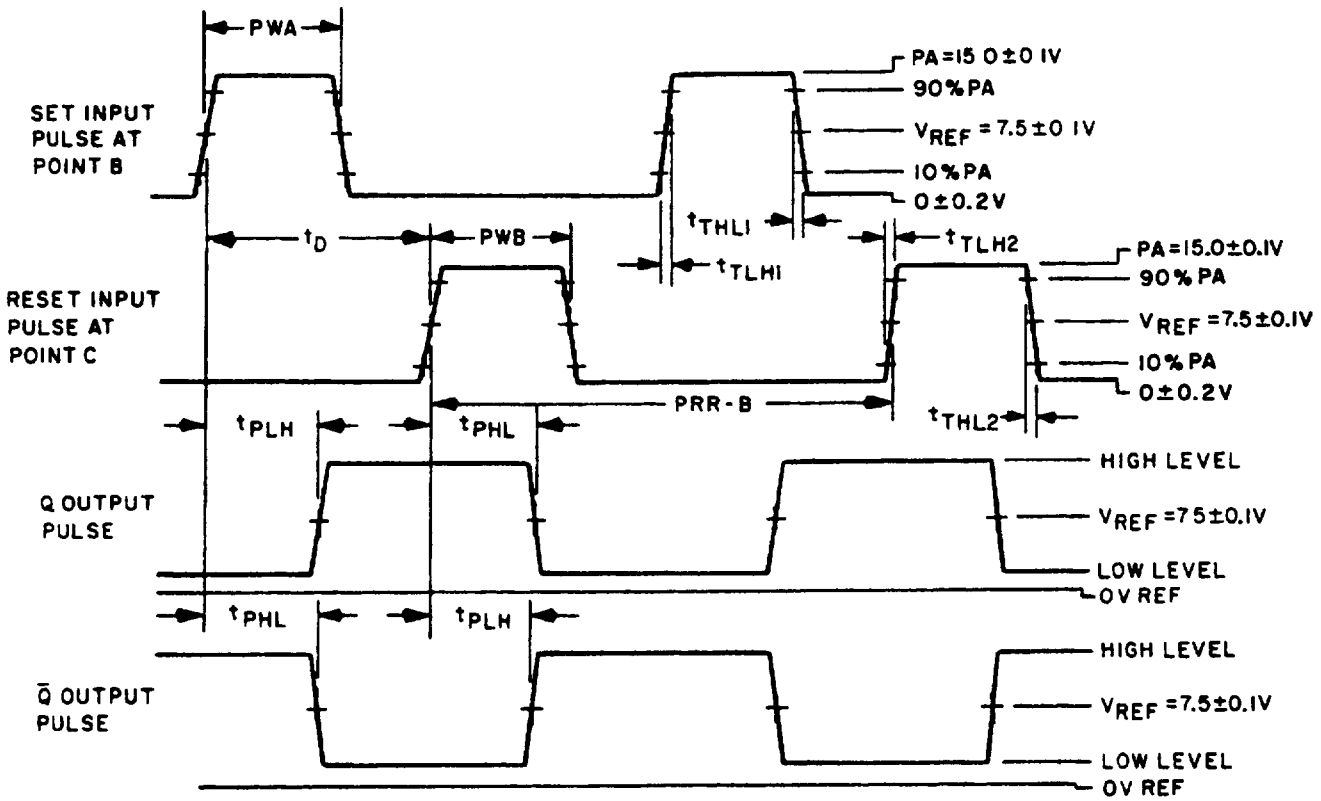
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NOTE C1 = 50 ±5 pF (including probe and parasitic capacitance)

FIGURE 27 Propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXXI and XXXII

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NOTES
1. Item J01-1 input pulse parameters
   PWA = 2.0 ±0.2 µs
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters
   PWB = 2.0 ±0.2 µs.
   PRR-B = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. t_D = 2.0 ±0.2 µs.
4. Before making the measurements, verify that the output pulse high level is ≥ 13.5 V and that the low level is ≤ 1.5 V
5. The lead length of electrical conductors between input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25 4 mm).
6. All voltages are referenced to ground

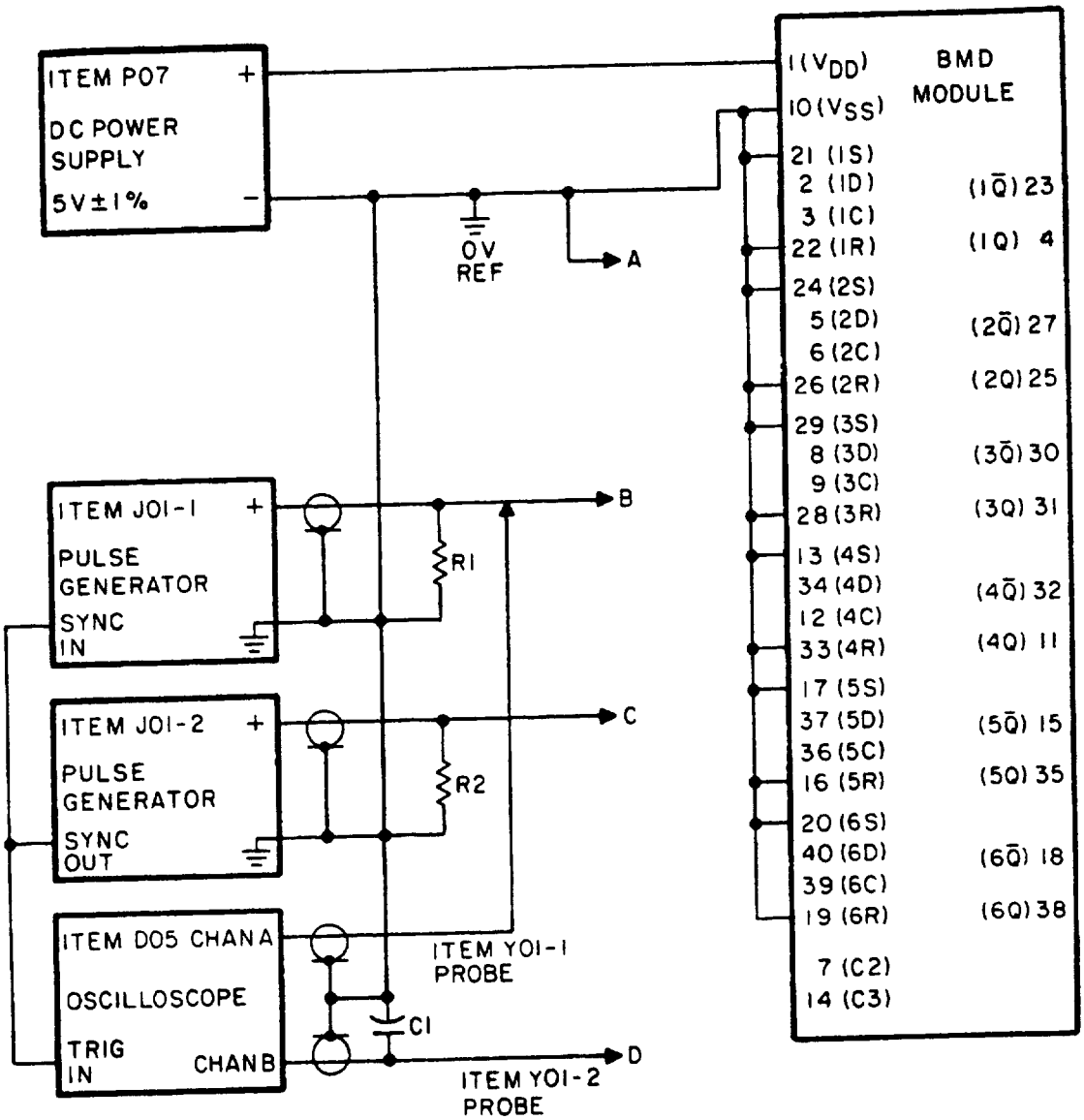
FIGURE 28 Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXXI and XXXII.

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TABLE XXXIII. Maximum shift frequency check test.

Test condition	Parameter		Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Max	Unit	Test figure
GG1	1Q	Maximum shift frequency check	5,6,8,9,34,12,37,36,40,39	3	2	4	900	1100	ns	29,30
GG2	1$\bar{Q}$	↓	5,6,8,9,34,12,37,36,40,39	3	2	23	↓	↓	↓	↓
GG3	2Q		2,3,8,9,34,12,37,36,40,39	6	5	25				
GG4	2$\bar{Q}$		2,3,8,9,34,12,37,36,40,39	6	5	27				
GG5	3Q		2,3,5,6,34,12,37,36,40,39	9	8	31				
GG6	3$\bar{Q}$		2,3,5,6,34,12,37,36,40,39	9	8	30				
GG7	4Q		2, 3,5,6,8,9,37,36,40,39	12	34	11				
GG8	4$\bar{Q}$		2,3,5,6,8,9,37,36,40,39	12	34	32				
GG9	5Q		2,3,5,6,8,9,12,34,40,39	36	37	35				
GG10	5$\bar{Q}$		2,3,5,6,8,9,12,34,40,39	36	37	15				
GG11	6Q		2,3,5,6,8,9,12,34,36,37	39	40	38				
GG12	6$\bar{Q}$		2,3,5,6,8,9,12,34,36,37	39	40	18				

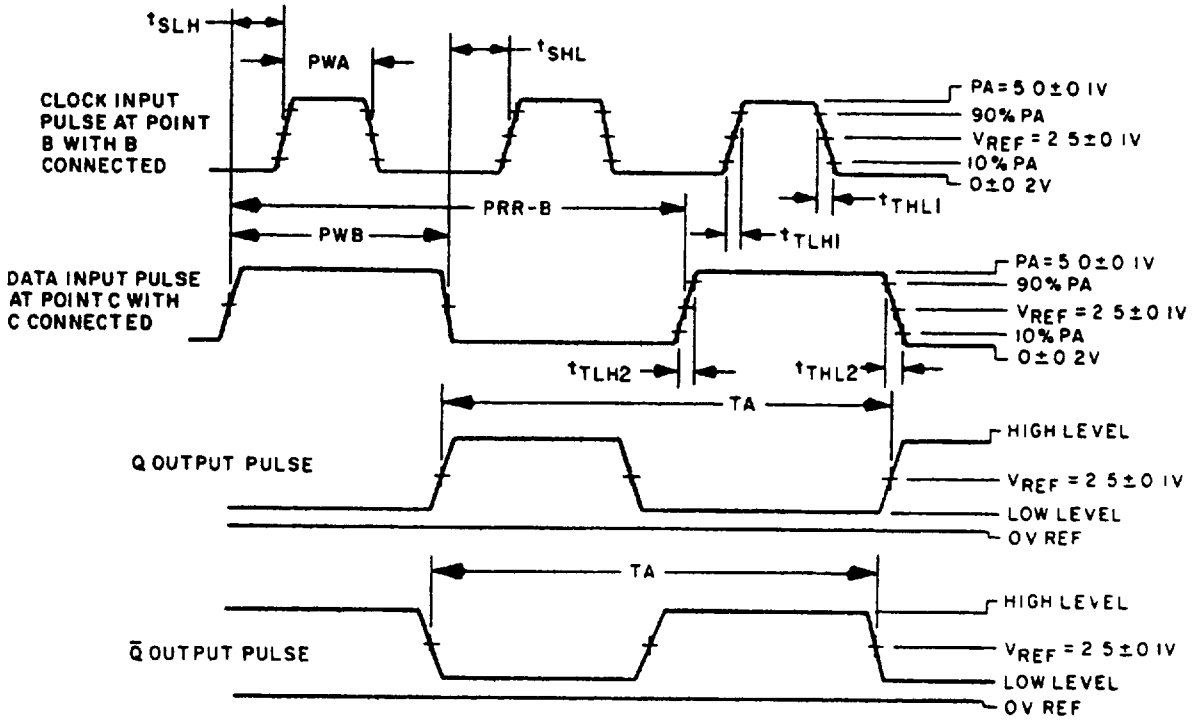
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:07Z
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NOTES
1 R1 = R2 = 51.1Ω ±5%, 1/2 W
2 C1 = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 29. Maximum shift frequency test circuit for table XXXIII.

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NOTES:
1. Item J01-1 input pulse parameters (item J01-1 in double pulse mode):
   PWA = 250 +25, -0 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWB = 500 ±50 ns
   PRR-B = 1.0 ±0.1 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns
3. $t_{SLH} = t_{SHL}$ = 100 ±10 ns
4. TA = 900 - 1100 ns (see table XXXIII initial limits column).
5. Before making any measurements, verify that the output pulse high level is $\geq$ 4.6 V and that the low level is $\leq$ 0.4 V.
6. The lead length of electrical conductors between terminator resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
7. All voltages are referenced to ground

FIGURE 30. Maximum shift frequency test circuit waveforms for table XXXIII

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TABLE XXXIV Power supply filter capacitance.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
HH	Power supply filter capacitance	Open	1	0.007	0.013	µF	31

TABLE XXXV. C2 and C3 capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
JJ1	C2 Capacitance	7	Open	176	264	pF	31
JJ2	C3 Capacitance	14	Open	176	264	pF	31

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TABLE XXXVI. Input capacitance test.

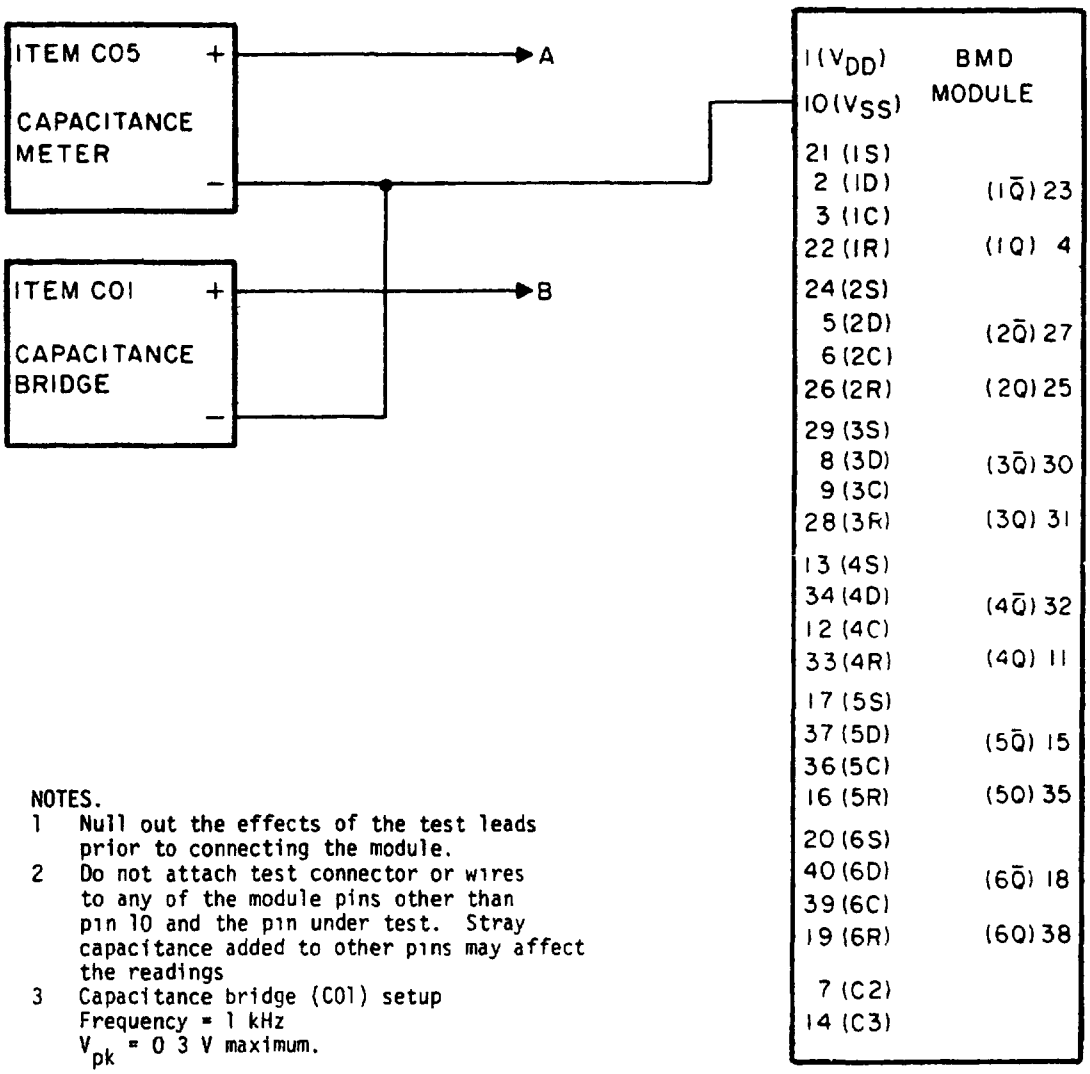
Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
KK1	1S Input capacitance	21	2	15	pF	31
KK2	1D	2				
KK3	1C	3				
KK4	1R	22				
KK5	2S	24				
KK6	2D	5				
KK7	2C	6				
KK8	2R	26				
KK9	3S	29				
KK10	3D	8				
KK11	3C	9				
KK12	3R	28				
KK13	4S	13				
KK14	4D	34				
KK15	4C	12				
KK16	4R	33				
KK17	5S	17				
KK18	5D	37				
KK19	5C	36				
		Open				

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TABLE XXXVI. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
					Min	Max		
KK20	5R	Input capacitance	16	Open	2	15	pF	31
KK21	6S	↓	20	↓	↓	↓	↓	↓
KK22	6D		40					
KK23	6C		39					
KK24	6R		19					

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NOTES.
1 Null out the effects of the test leads prior to connecting the module.
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3 Capacitance bridge (CO1) setup
Frequency = 1 kHz
V_{pk} = 0 3 V maximum.

FIGURE 31. Power supply filter, C2 and C3, and input capacitance test circuit for tables XXXIV, XXXV and XXXVI

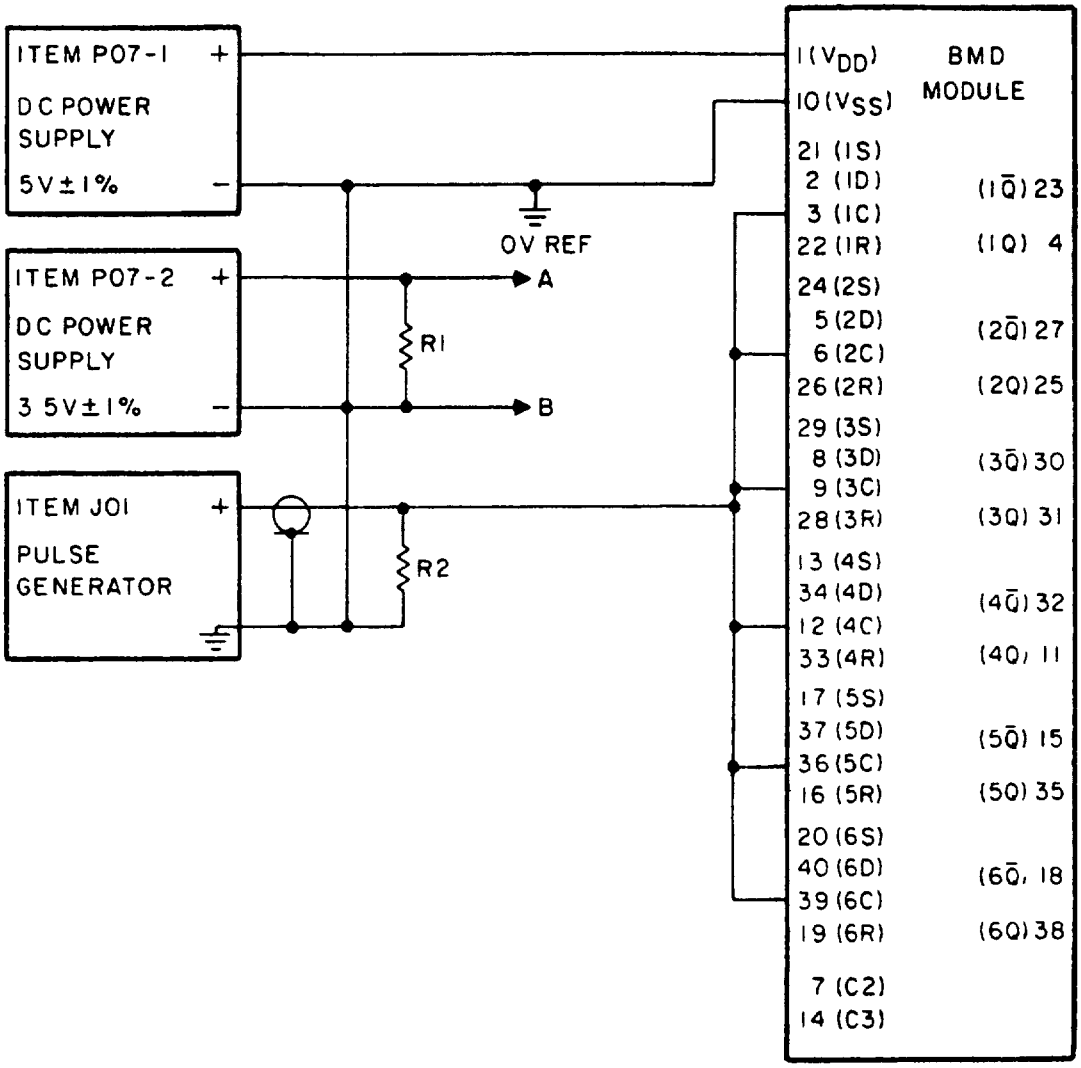
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TABLE XXXVII. Functional test.

Test condition	Parameter	Connect		Outputs												Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	1Q	2Q	3Q	4Q	5Q	6Q	$\overline{1Q}$	$\overline{2Q}$	$\overline{3Q}$	$\overline{4Q}$	$\overline{5Q}$	$\overline{6Q}$	Min			
LL1	Functional test	Open	21,2,22,24,5,26,29,8,28,13,34,33,17,37,16,20,40,19	L	L	L	L	L	L	H	H	H	H	H	H	1/	1/	---	32
LL2	Functional test	2,5,8,34,37,40	21,22,24,26,29,28,13,33,17,16,20,19	H	H	H	H	H	H	L	L	L	L	L	L	1/	1/	---	32
LL3	Functional test	21,22,24,26,29,28,13,33,17,16,20,19	2,5,8,34,37,40,3,6,9,12,36,39	H	H	H	H	H	H	H	H	H	H	H	H	1/	1/	---	32

1/ L = 0.0 V min, 0.4 V max
H = 4.6 V min, 5.0 V max

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NOTES.
1. R1 = R2 = 50Ω ±5%, 1/2 W
2. Item J01 pulse parameters
   PW = 1.0 µs ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
   PA = 5.0 ±0 1 V.
3 For test condition LL3, disconnect item J01 before making measurements

FIGURE 32. Functional test circuit for table XXXVII

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TABLE XXXVIII. Data pulse set time (t_{SLH} and t_{SHL}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
MM1	1Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
MM2	1$\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
MM3	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
MM4	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
MM5	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
MM6	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
MM7	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
MM8	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
MM9	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
MM10	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13 33,17,37,36,16,20,40,39,19	36	37	15				
MM11	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
MM12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XXXVIII. Data pulse set time (t_{SLH} and t_{SHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
MM13	1Q Data pulse set time (t_{SHL}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
MM14	1$\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
MM15	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
MM16	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
MM17	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
MM18	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
MM19	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
MM20	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
MM21	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
MM22	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
MM23	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
MM24	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XXXIX. Data pulse hold time (t_{HHL} and t_{HLH}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
NN1	1Q Data pulse hold time (t_{HHL}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
NN2	1$\bar{Q}$	21,2,3,22,24,29,6,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
NN3	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
NN4	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
NN5	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
NN6	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,26,13, 34,12,33,17,37,36,16,20,19	9	8	30				
NN7	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
NN8	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
NN9	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
NN10	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
NN11	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
NN12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XXXIX. Data pulse hold time (t_{HHL} and t_{HLH}) test - Continued.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
NN13	1Q Data pulse hold time (t_{HLH}), C_L = 50 pF, V_{DD} = 5 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
NN14	1$\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
NN15	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
NN16	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
NN17	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
NN18	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
NN19	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
NN20	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
NN21	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
NN22	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
NN23	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
NN24	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XL. Data pulse set time (t_{SLH} and t_{SHL}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
PP1	1Q Data pulse set time (t_{SLH}), C_L = 50 pf, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
PP2	1$\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
PP3	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
PP4	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
PP5	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
PP6	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
PP7	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
PP8	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
PP9	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
PP10	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
PP11	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
PP12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XL. Data pulse set time (t_{SLH} and t_{SHL}) test - Continued.

Test condition		Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
PP13	1Q	Data pulse set time (t_{SHL}), C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
PP14	1$\bar{Q}$		21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
PP15	2Q		21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
PP16	2$\bar{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
PP17	3Q		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
PP18	3$\bar{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
PP19	4Q		21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
PP20	4$\bar{Q}$		21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
PP21	5Q		21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
PP22	5$\bar{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
PP23	6Q		21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
PP24	6$\bar{Q}$		21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XLI. Data pulse hold time (t_{HHL} and t_{HLH}) test.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
QQ1	1Q Data pulse hold time (t_{HHL}). C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	us	33,34
QQ2	1$\overline{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
QQ3	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
QQ4	2$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
QQ5	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
QQ6	3$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
QQ7	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
QQ8	4$\overline{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
QQ9	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
QQ10	5$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
QQ11	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
QQ12	6$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XLI. Data pulse hold time (t_{HHL} and t_{HLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
QQ13	1Q Data pulse hold time (t_{HLH}), C_L = 50 pF, V_{DD} = 10 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
QQ14	1$\overline{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
QQ15	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
QQ16	2$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
QQ17	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
QQ18	3$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
QQ19	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
QQ20	4$\overline{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
QQ21	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
QQ22	5$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
QQ23	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
QQ24	6$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XLII. Data pulse set time (t_{SLH} and t_{SHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
RR1	1Q Data pulse set time (t_{SLH}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
RR2	1$\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
RR3	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
RR4	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
RR5	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
RR6	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
RR7	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
RR8	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
RR9	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
RR10	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
RR11	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
RR12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XLII. Data pulse set time (t_{SLH} and t_{SHL}) test - Continued.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
RR13	1Q Data pulse set time (t_{SHL}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
RR14	1$\bar{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
RR15	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
RR16	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
RR17	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
RR18	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
RR19	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
RR20	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
RR21	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
RR22	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
RR23	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
RR24	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XLIII Data pulse hold time (t_{1H1H} and t_{1H1L}) test

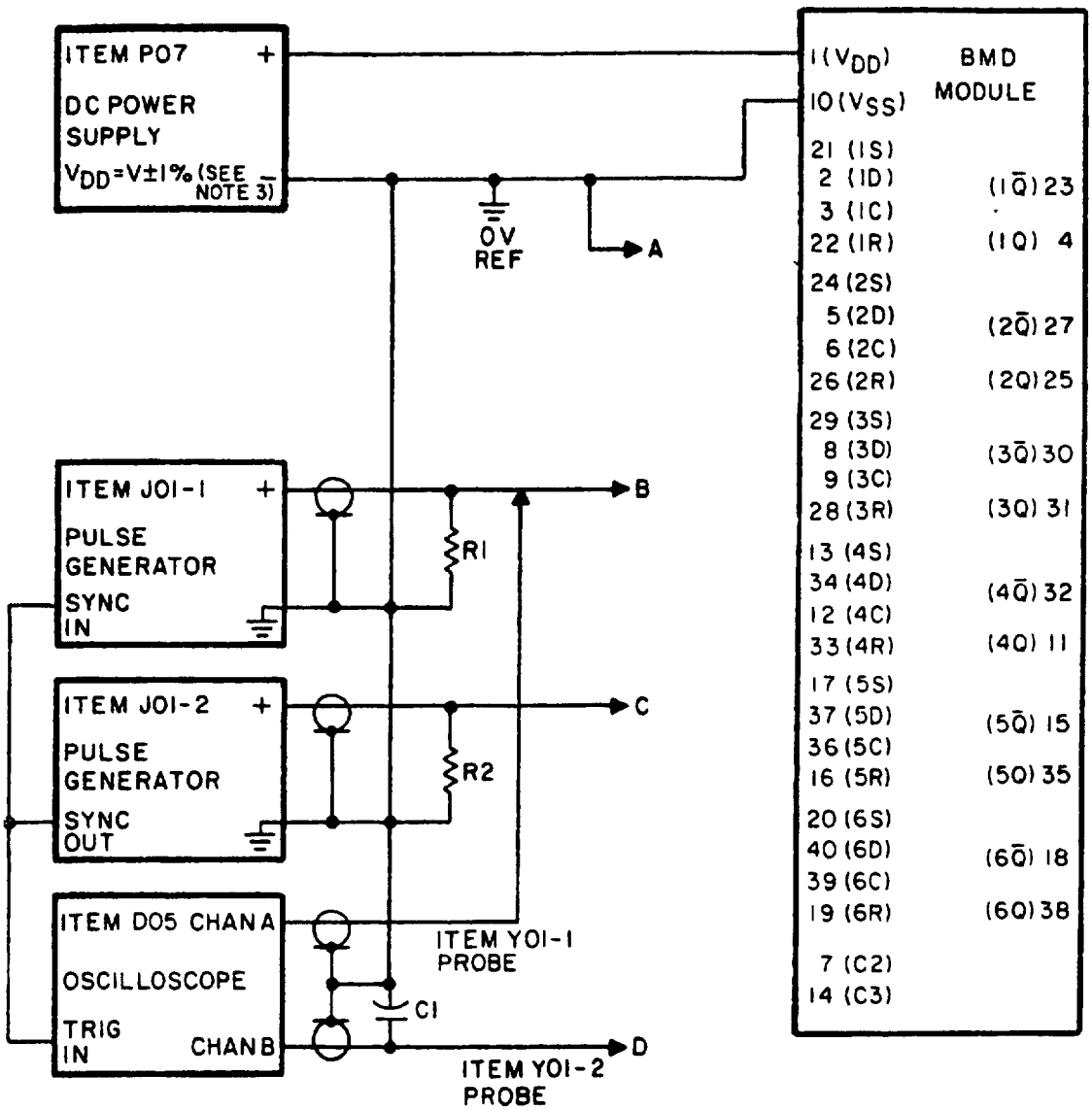
Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
SS1	1Q Data pulse hold time (t_{HLH}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
SS2	1$\bar{Q}$	21,2,3,22,24,29,8,9,26,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
SS3	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
SS4	2$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
SS5	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
SS6	3$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
SS7	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
SS8	4$\bar{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
SS9	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
SS10	5$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
SS11	6Q	21,2,3,22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
SS12	6$\bar{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

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TABLE XLIII. Data pulse hold time (t_{HLH} and t_{HHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
SS13	1Q Data pulse hold time (t_{HHL}), C_L = 50 pF, V_{DD} = 15 V	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	3	2	4	3.6	4.4	µs	33,34
SS14	1$\overline{Q}$	21,2,3,22,24,29,8,9,28,13,34,12, 33,17,37,36,16,20,40,39,19	3	2	23				
SS15	2Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	6	5	25				
SS16	2$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	6	5	27				
SS17	3Q	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,16,20,40,39,19	9	8	31				
SS18	3$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	9	8	30				
SS19	4Q	21,22,24,5,6,26,29,8,9,28,13,34, 12,33,17,37,36,20,40,39,19,16	12	34	11				
SS20	4$\overline{Q}$	21,2,3,22,24,26,29,8,9,28,13,34, 12,33,17,37,36,16,20,40,39,19	12	34	32				
SS21	5Q	21,2,3,22,24,5,6,26,29,28,13,34, 12,33,17,37,36,16,20,40,39,19	36	37	35				
SS22	5$\overline{Q}$	21,2,3,22,24,5,6,26,29,8,9,28,13, 33,17,37,36,16,20,40,39,19	36	37	15				
SS23	6Q	21,[illegible],22,24,5,6,29,8,9,28,13,34, 12,33,17,16,20,40,39,19	39	40	38				
SS24	6$\overline{Q}$	21,2,3,[illegible],24,5,6,26,29,8,9,28,13, 34,12,33,17,37,36,16,20,19	39	40	18				

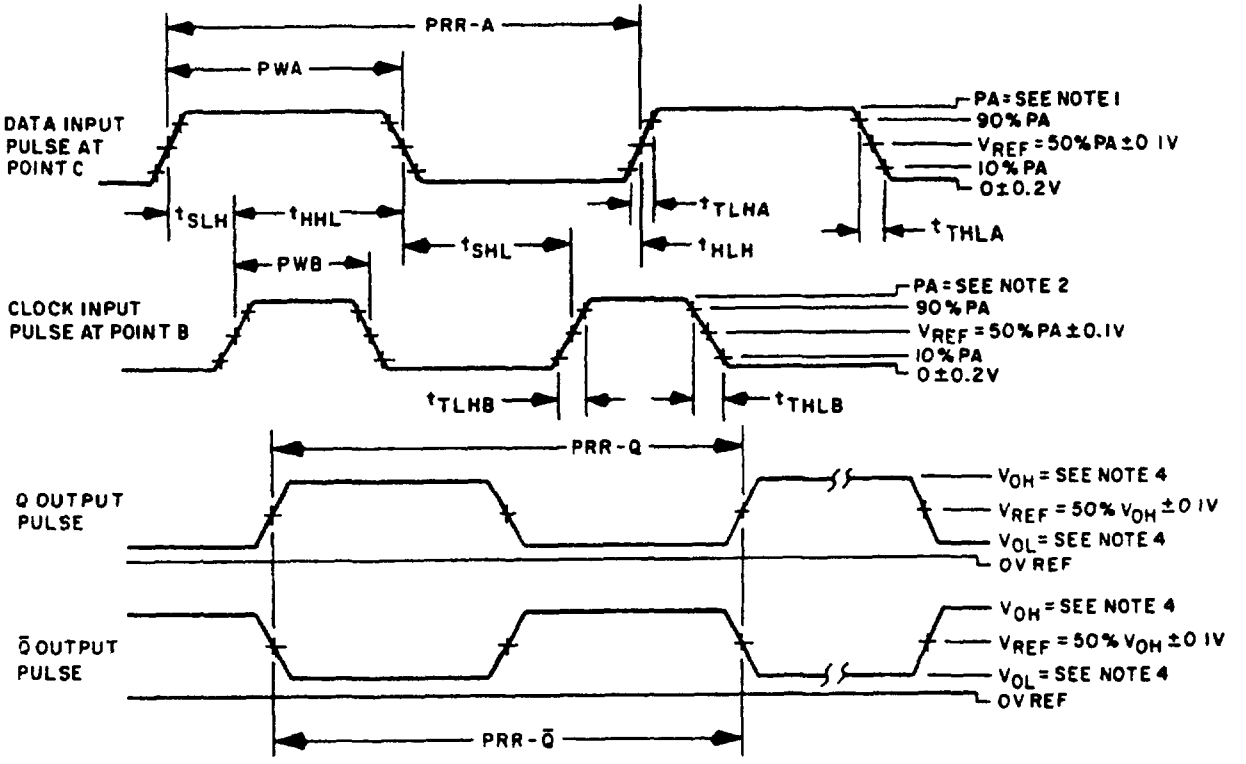
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NOTES.
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. C1 = 50 ±5 pF (including probe and parasitic capacitance).
3. V_{DD} = 5 V for test conditions MM1-MM24, NN1-NN24
   = 10 V for test conditions PP1-PP24, QQ1-QQ24.
   = 15 V for test conditions RR1-RR24, SS1-SS24.

FIGURE 33. Data pulse set time (t_{SLH} and t_{SHL}) and data pulse hold time (t_{HHL} and t_{HLH}) test circuit for tables XXXVIII through XLIII.

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NOTES:
1. Data input pulse parameters
   PWA = 2.0 ±0.2 µs
   PRR-A = 4.0 ±0.4 µs
   $t_{TLHA} = t_{THLA}$ = 10 ±2 ns
   PA = See table XLIV
2. Clock input pulse parameters (item 101-1 in double pulse mode):
   PWB = 1.0 ±0.1 µs.
   $t_{TLHB} = t_{THLB}$ = 10 ±2 ns
   PA = See table XLIV
3. Set t_{SLH}, t_{SHL}, t_{HHL} and t_{HLH} per table XLIV.
4. Verify that output levels are:
   V_{DD} = 5 V, $V_{OL} \leq 0.4$ V, $V_{OH} \geq 4.6$ V
   V_{DD} = 10 V, $V_{OL} \leq 0.5$ V, $V_{OH} \geq 9.5$ V
   V_{DD} = 15 V, $V_{OL} \leq 1.5$ V, $V_{OH} \geq 13.5$ V.

FIGURE 34. Data pulse set time (t_{SLH} and t_{SHL}) and data pulse hold time (t_{HHL} and t_{HLH}) test circuit waveforms for tables XXXVIII through XLIII.

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TABLE VIII. Data pulse set time (t_{SLH} and t_{SHL}) and data pulse hold time (t_{HHL} and t_{HLH}) test parameters for figure 34.

Test condition	V_{DD}	PA	t_{SLH}	t_{SHL}	T_{HHL}	t_{HLH}
MM1-MM12	5 V	5.0 ±0.1 V	40 +4,-0 ns	100 +10,-0 ns		
MM13-MM24	5 V	5.0 ±0.1 V	100 +10,-0 ns	40 +4,-0 ns		
NN1-NN12	5 V	5.0 ±0.1 V			40 +4,-0 ns	100 +10,-0 ns
NN13-NN24	5 V	5.0 ±0.1 V			100 +10,-0 ns	40 +4,-0 ns
PP1-PP12	10 V	10.0 ±0.1 V	30 +3,-0 ns	100 +10,-0 ns		
PP13-PP24	10 V	10.0 ±0.1 V	100 +10,-0 ns	30 +3,-0 ns		
QQ1-QQ12	10 V	10.0 ±0.1 V			30 +3,-0 ns	100 +10,-0 ns
QQ13-QQ24	10 V	10.0 ±0.1 V			100 +10,-0 ns	30 +3,-0 ns
RR1-RR12	15 V	15.0 ±0.1 V	25 +2,-0 ns	100 +10,-0 ns		
RR13-RR24	15 V	15.0 ±0.1 V	100 +10,-0 ns	25 +2,-0 ns		
SS1-SS12	15 V	15.0 ±0.1 V			25 +2,-0 ns	100 +10,-0 ns
SS13-SS24	15 V	15.0 ±0.1 V			100 +10,-0 ns	25 +2,-0 ns

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